

AI 500

Foundations of Artificial Intelligence

L5: Unsupervised Learning

By Murtaza Taj



Unsupervised Learning



- Clustering
 - **Kmeans/Kmeans++**
 - **K-mediod**
 - **DBSCAN**
 - GMM
- Hierarchical Clustering
 - Agglomerative Clustering
 - Divisive Clustering (Isolated Forest)

Density Based Clustering

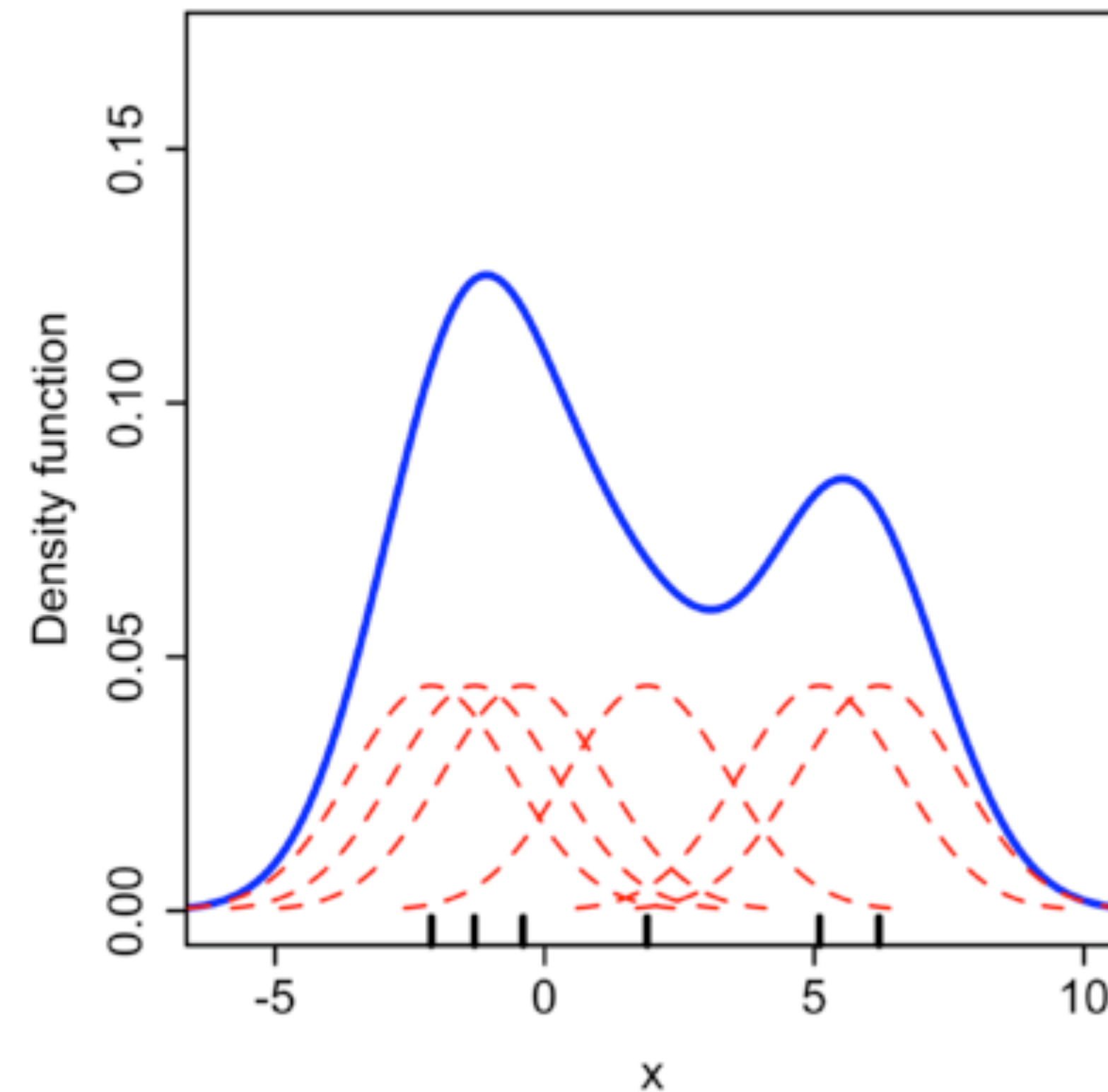
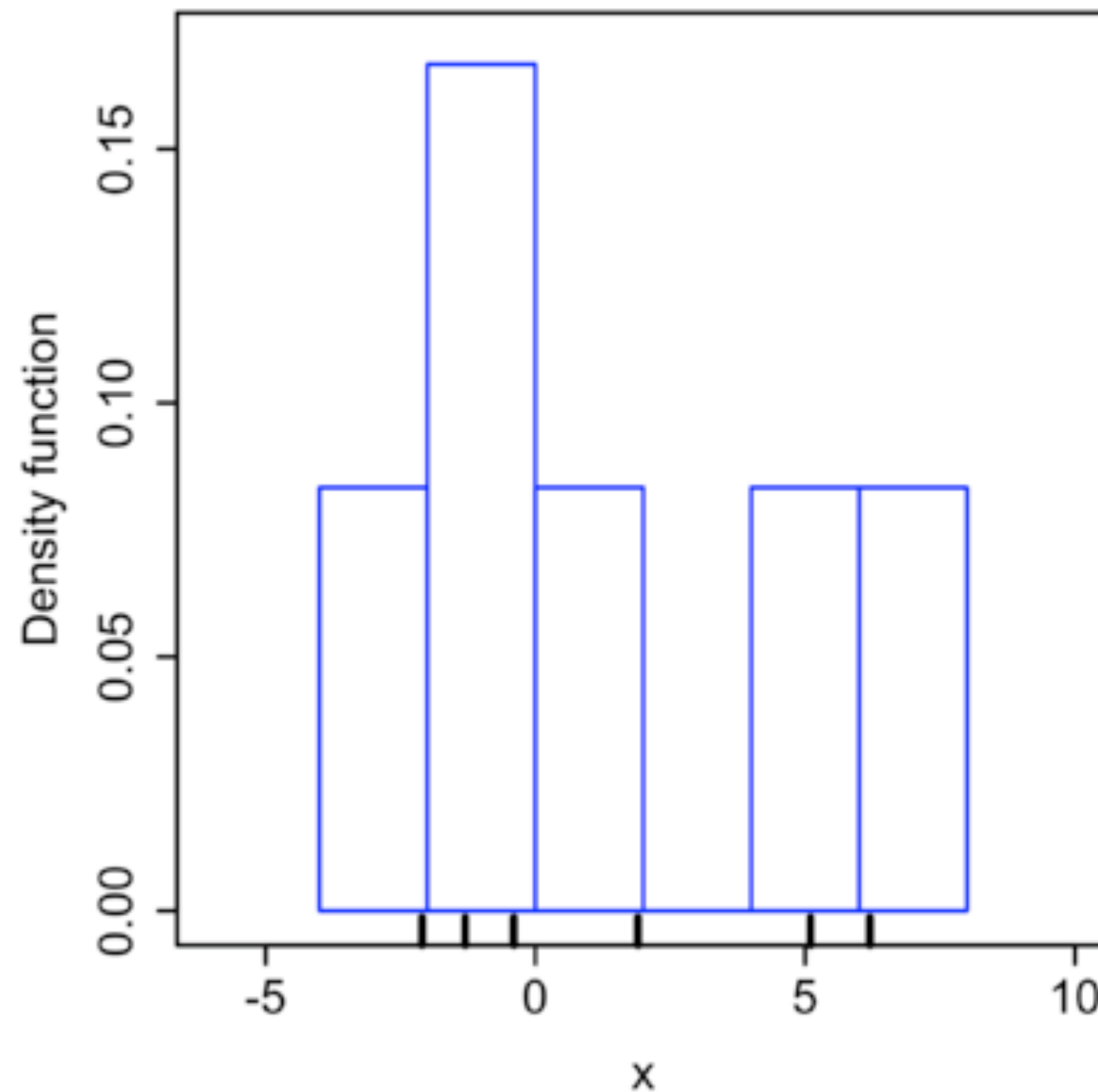
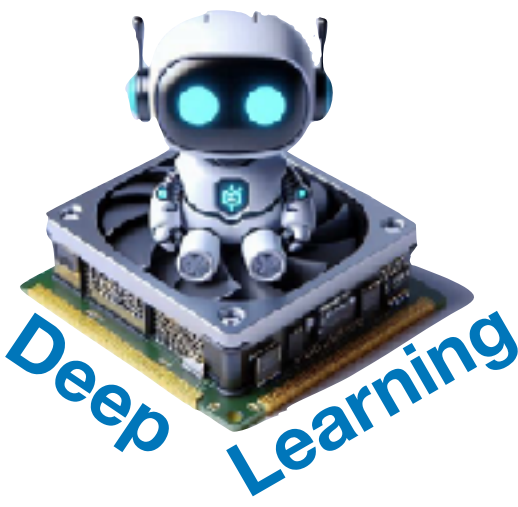
KDE & GMM

Kernel Density Estimation (KDE)

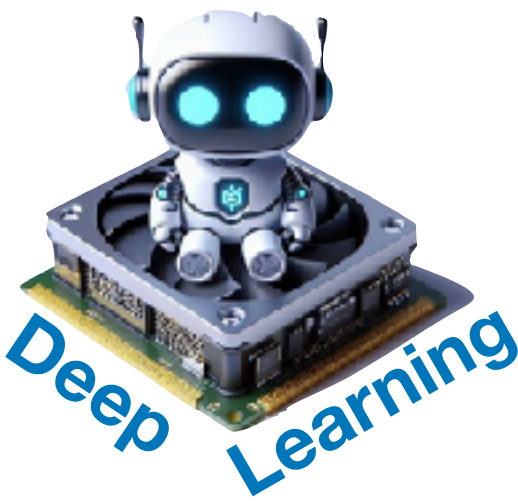
Parzen window

- $x_1 = -2.1, x_2 = -1.3, x_3 = -0.4, x_4 = 1.9, x_5 = 5.1, x_6 = 6.2$
- 6 bins each of width 2
- $[-4 -2], [-2 0], [0 2], [2 4], [4 6], [6 8]$

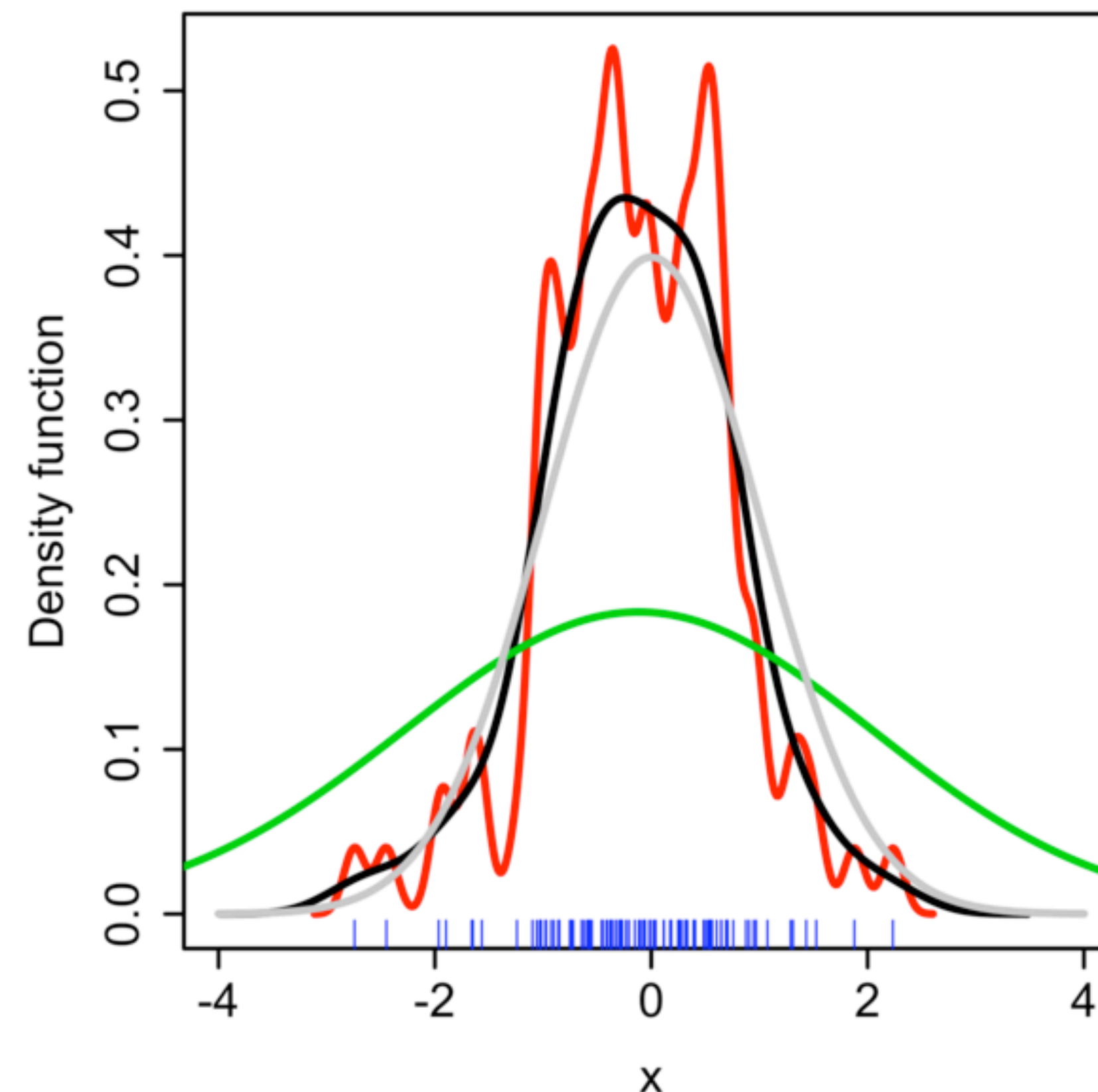
$$f(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$



Bandwidth selection

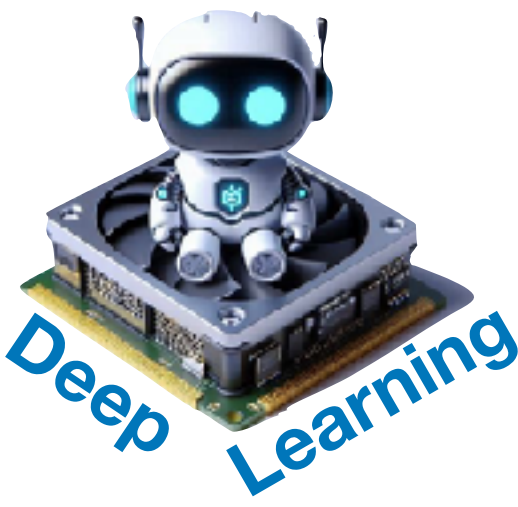


- Kernel density estimate (KDE) with different bandwidths of a random sample of 100 points from a standard normal distribution.
- Grey: true density (standard normal).
- Red: KDE with $h=0.05$. Green: KDE with $h=2$. Black: KDE with $h=0.337$.



KDE Example

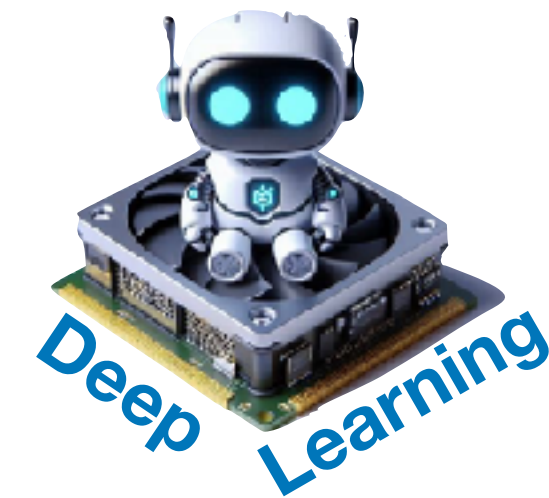
Excel Sheet



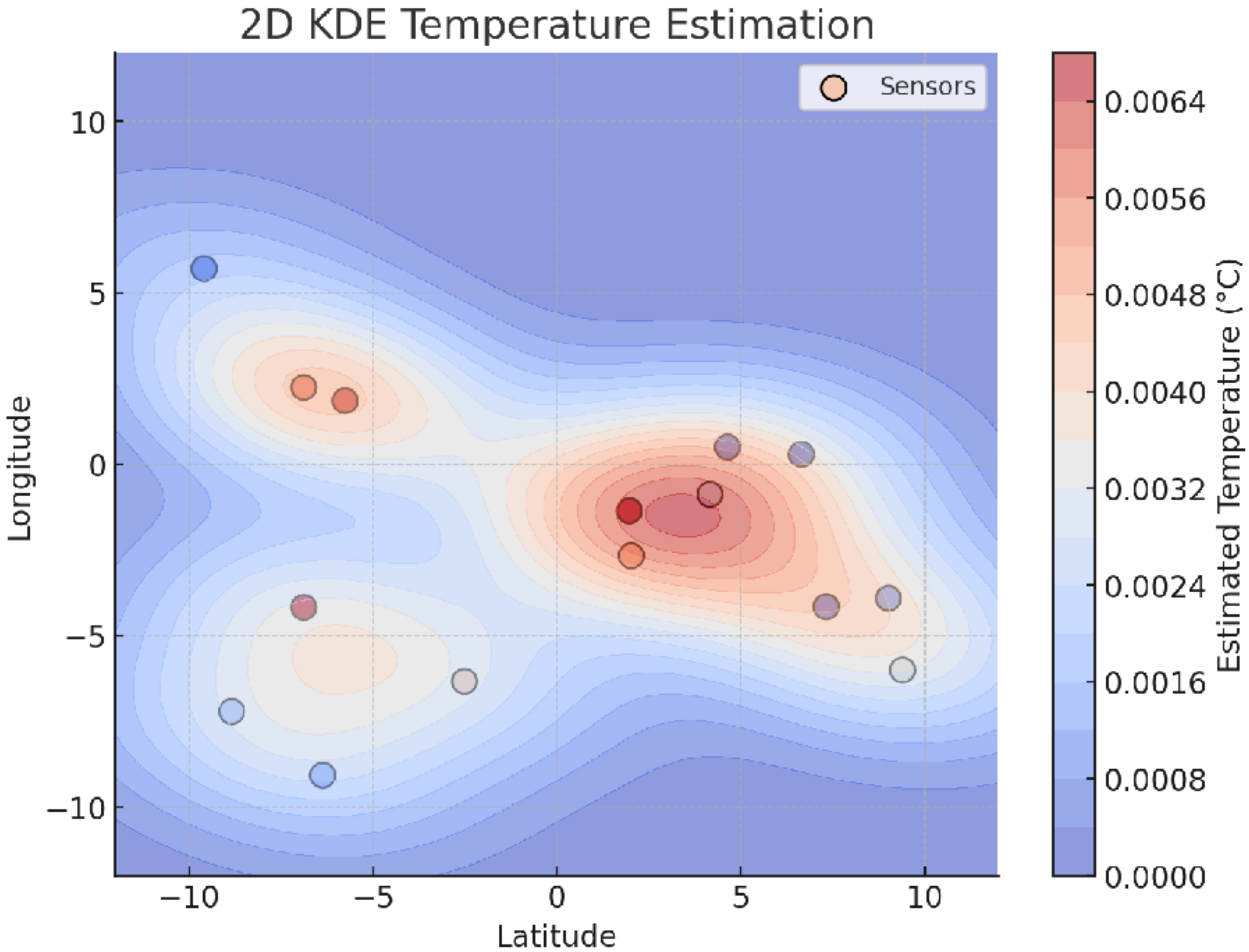
$$f(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

KDE Example

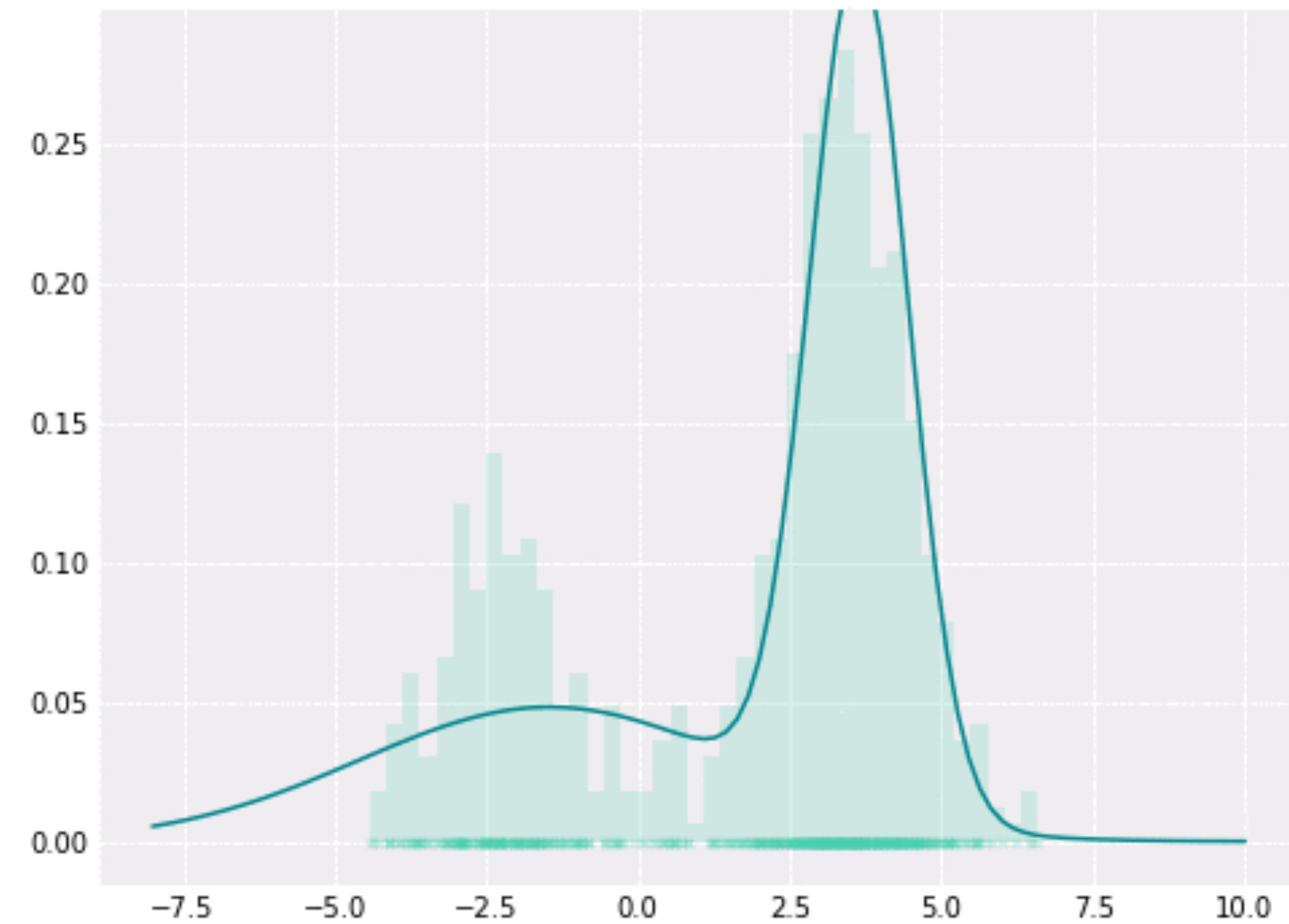
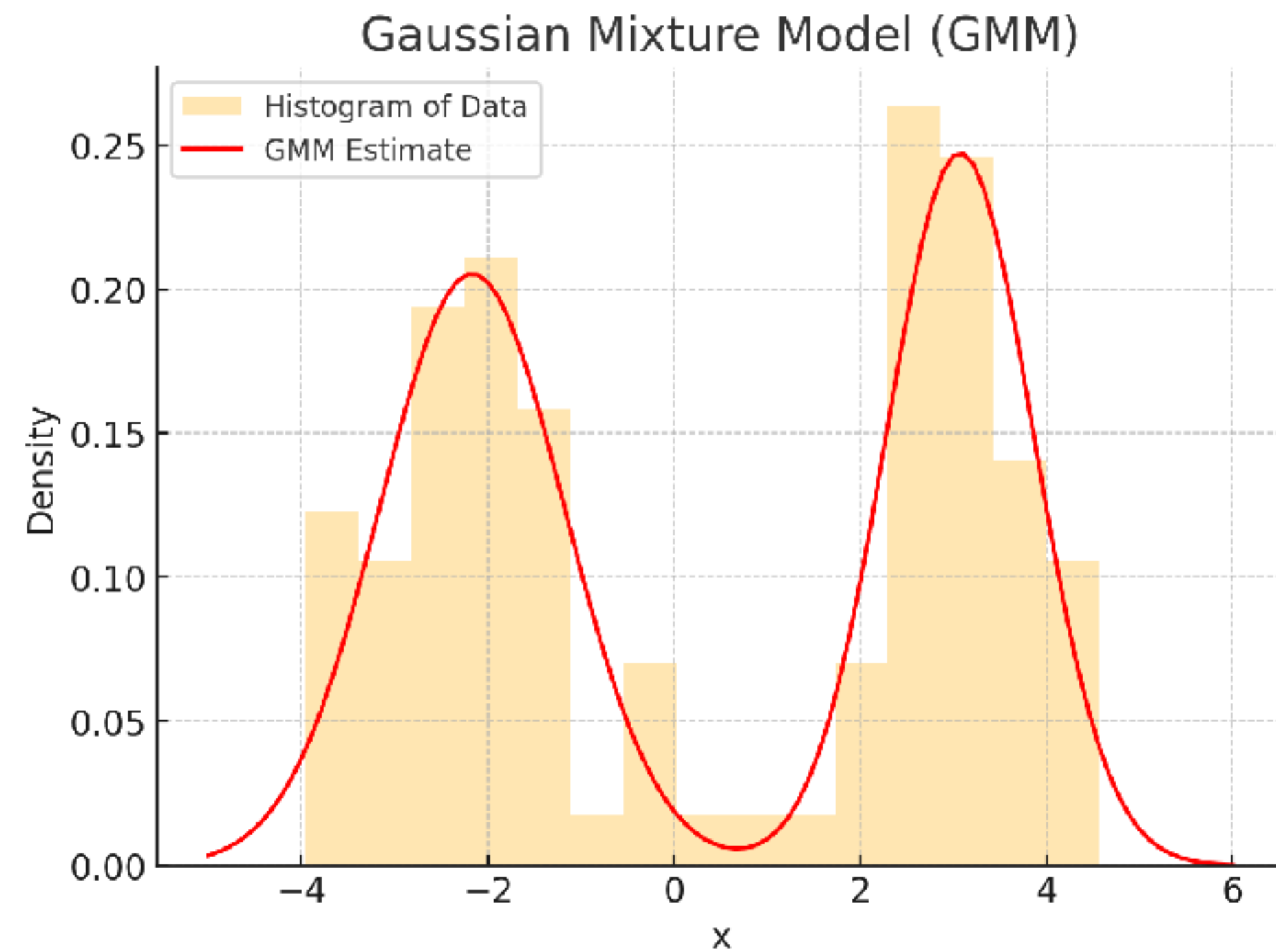
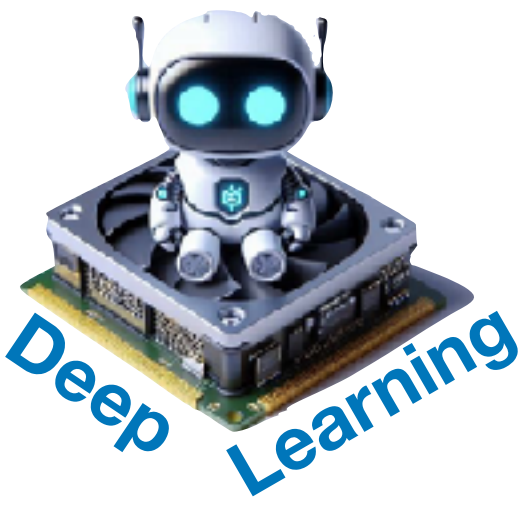
Temperature Estimation



Sensor ID	Latitude	Longitude	Temperature (°C)
1	-2.50920	-6.33191	27.15
2	9.01429	-3.91516	18.41
3	4.63988	0.49513	16.30
4	1.97317	-1.36110	33.98
5	-6.87963	-4.17542	34.31
6	-6.88011	2.23706	31.17
7	-8.83833	-7.21012	21.09
8	7.32352	-4.15711	16.95
9	2.02230	-2.67276	28.68
10	4.16145	-0.87860	23.80
11	-9.58831	5.70352	17.44
12	9.39820	-6.00652	24.90
13	6.64885	0.28469	15.69
14	-5.75322	1.84829	33.19
15	-6.36350	-9.07099	20.18



Gaussian Mixture Model (GMM)



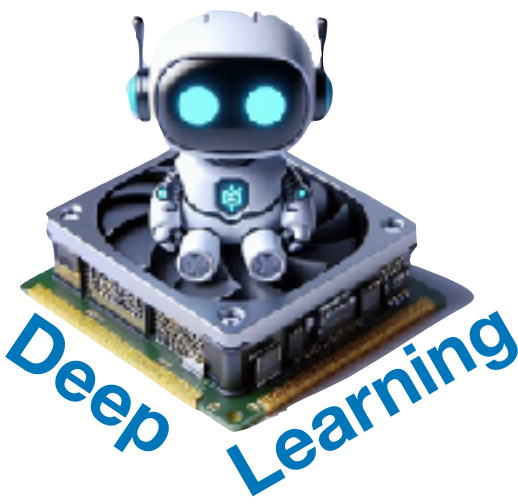
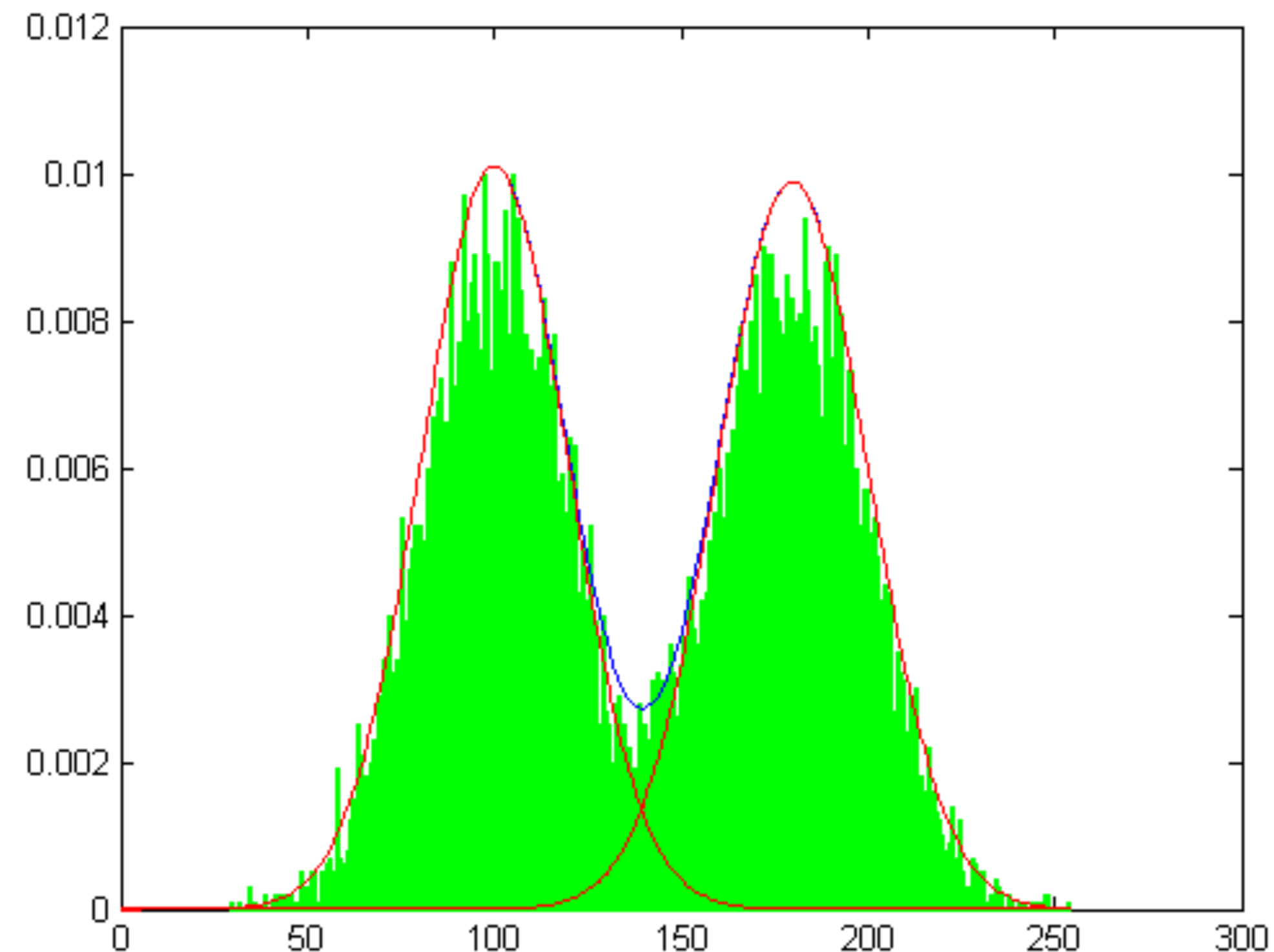
Deom: <https://lukapopijac.github.io/gaussian-mixture-model/>

Gaussian Mixture Model (GMM)

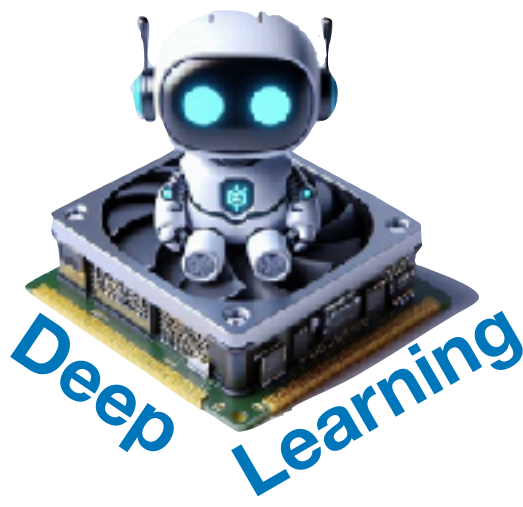
2 Gaussians only

$$\mu_1 = 100, \sigma_1 = 20, \mu_2 = 180, \sigma_2 = 20, w_1 = 0.5$$

$$g(x) = 0.5 \times \left(\frac{1}{\sqrt{2\pi} (20)} e^{-\frac{(x-100)^2}{2(20)^2}} \right) + 0.5 \times \left(\frac{1}{\sqrt{2\pi} (20)} e^{-\frac{(x-180)^2}{2(20)^2}} \right)$$



Computing Parameters of the Mixture of Gaussians

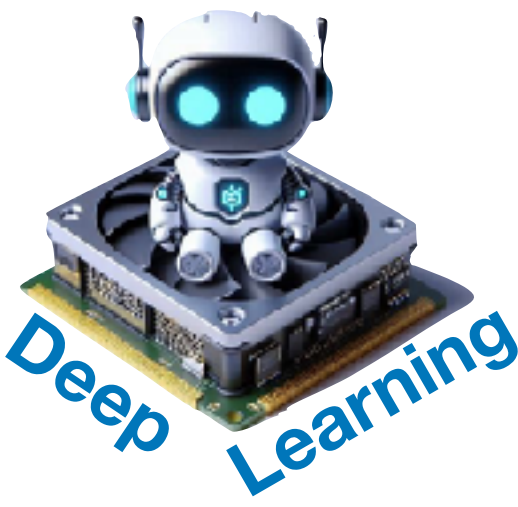


- We need to estimate the parameters of the Mixture of Gaussians *pdf* that fits the data

$$g(x | w_1, \mu_1, \sigma_1, \mu_2, \sigma_2) = \frac{w_1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} + \frac{1-w_1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x-\mu_2)^2}{2\sigma_2^2}}$$

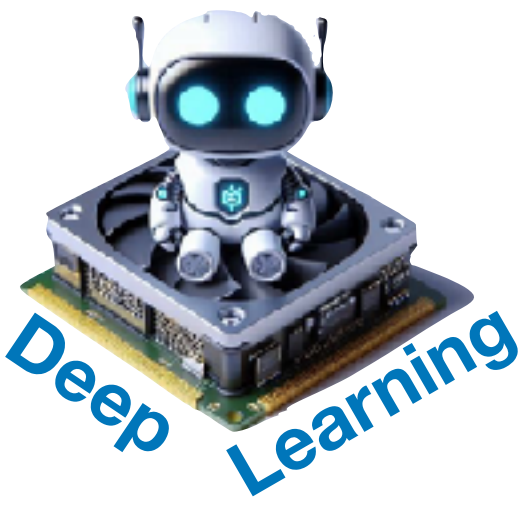
- Hard problem, because assignment of data in the mixture is not known
 - We do not know which histogram bin belongs to which Gaussian distribution
- Use EM Algorithm
 - General algorithm for estimating mixtures

Expectation Maximization



- E-Step
 - Assume that we know the parameters
 - Then, finding the assignment of which point belongs to foreground or background is easy
- M-Step
 - Then assume that we know the assignment
 - Find the parameters
- Iterate

Expectation Maximization



- Initialize $w_1 \mu_1 \sigma_1 \mu_2 \sigma_2$
 - x is a vector for e.g. containing all the pixel values
 - $|X|$ is the length of x or number of pixels
- E-Step
 - Now we know the parameters, find how likely is each point to belong to each of the Gaussian distributions

x
100x100
 $|X| = 10000$

$$L_i(x) = w_i g(x | \mu_i, \sigma_i)$$

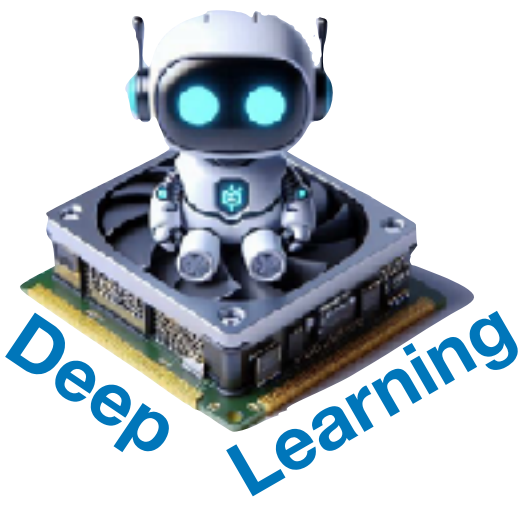
$$\text{where } g(x | \mu_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}$$

$$S_x = \sum_{i=1}^2 w_i g(x | \mu_i, \sigma_i) = \sum_{i=1}^2 L_i(x) \quad P_i(x) = \frac{L_i(x)}{S_x} = \frac{L_i(x)}{\sum_{j=1}^2 L_j(x)}$$

S_x 1x10,000

$P_i(x)$ 2x10,000

Expectation Maximization: M-Step



- Now that we know the contribution of each point to each distribution, **update** the parameters of the distribution

- Update of weight

$$w'_i = \frac{\sum_{x \in X} P_i(x)}{|\mathbf{X}|}$$

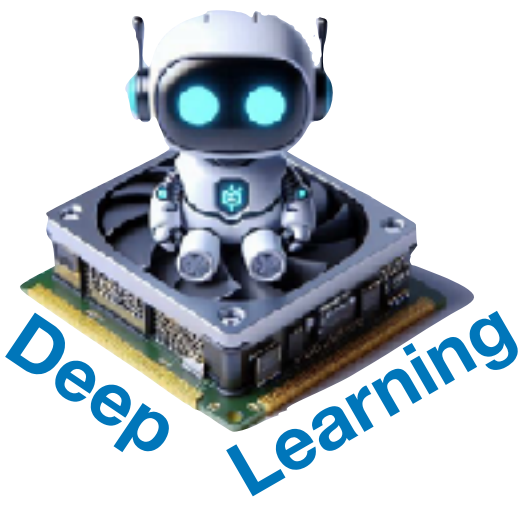
- Update of mean

$$\mu'_i = \frac{\sum_{x \in X} P_i(x)x}{\sum_{x \in X} P_i(x)}$$

- Update of variance

$$\sigma'^2_i = \frac{1}{\sum_{x \in X} P_i(x)} \sum_{x \in X} P_i(x) [(x - \mu'_i)^2]$$

Expectation Maximization - Histogram



- Initialize $w_1 \mu_1 \sigma_1 \mu_2 \sigma_2$

- x is a vector for e.g. containing all the pixel values
- $|X|$ is the length of x or number of pixels

x
100x100

$|X| = 10000$

- E-Step

- Now we know the parameters, find how likely is each point to belong to each of the Gaussian distributions

$$L_i(x) = w_i g(x | \mu_i, \sigma_i)$$

$$\text{where } g(x | \mu_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}$$

$$s_x = \sum_{i=1}^2 w_i g(x | \mu_i, \sigma_i) = \sum_{i=1}^2 L_i(x) \quad P_i(x) = \frac{L_i(x)}{s_x} = \frac{L_i(x)}{\sum_{j=1}^2 L_j(x)}$$

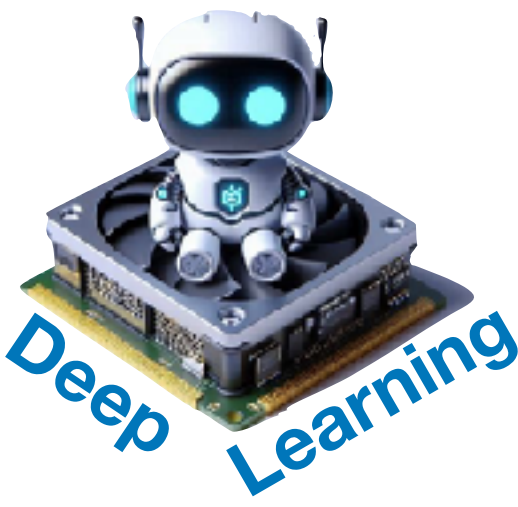
s_x

1x256

$P_i(x)$

2x256

Expectation Maximization -Histogram: M-Step



- Now that we know the contribution of each point to each distribution, **update** the parameters of the distribution

- Update of weight

$$w'_i = \frac{\sum_{x \in X} \hat{P}_i(x)}{|X|}$$

$$\hat{P}_i(x) = P_i(x)H(x)$$

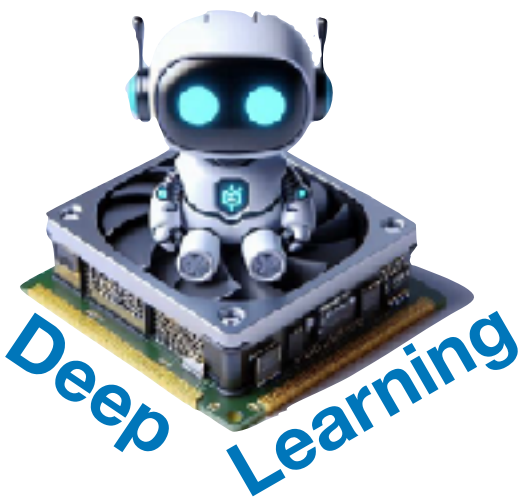
- Update of mean

$$\mu'_i = \frac{\sum_{x \in X} \hat{P}_i(x)x}{\sum_{x \in X} \hat{P}_i(x)}$$

- Update of variance

$$\sigma'^2_i = \frac{1}{\sum_{x \in X} \hat{P}_i(x)} \sum_{x \in X} \hat{P}_i(x) [(x - \mu'_i)^2]$$

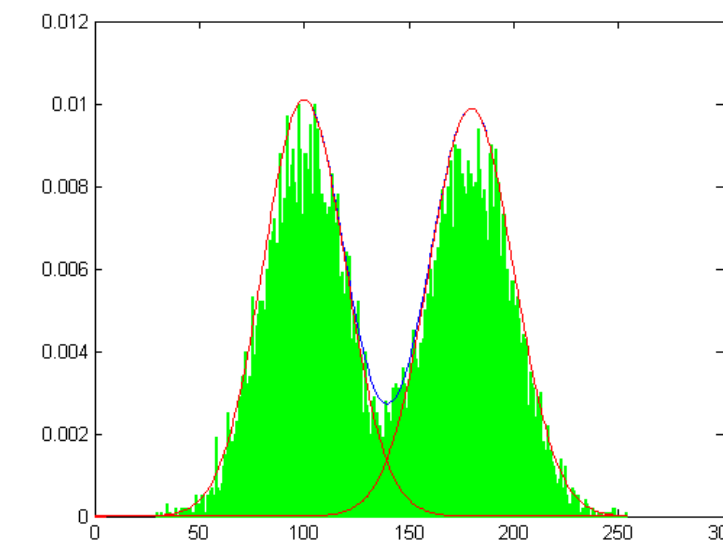
Finding the Threshold



- Now that the mixture parameters are known, the threshold is chosen to be the point where the two distributions intersect

- Threshold is the value of x where:

$$w_1 g(x | \mu_1, \sigma_1) = (1 - w_1) g(x | \mu_2, \sigma_2)$$

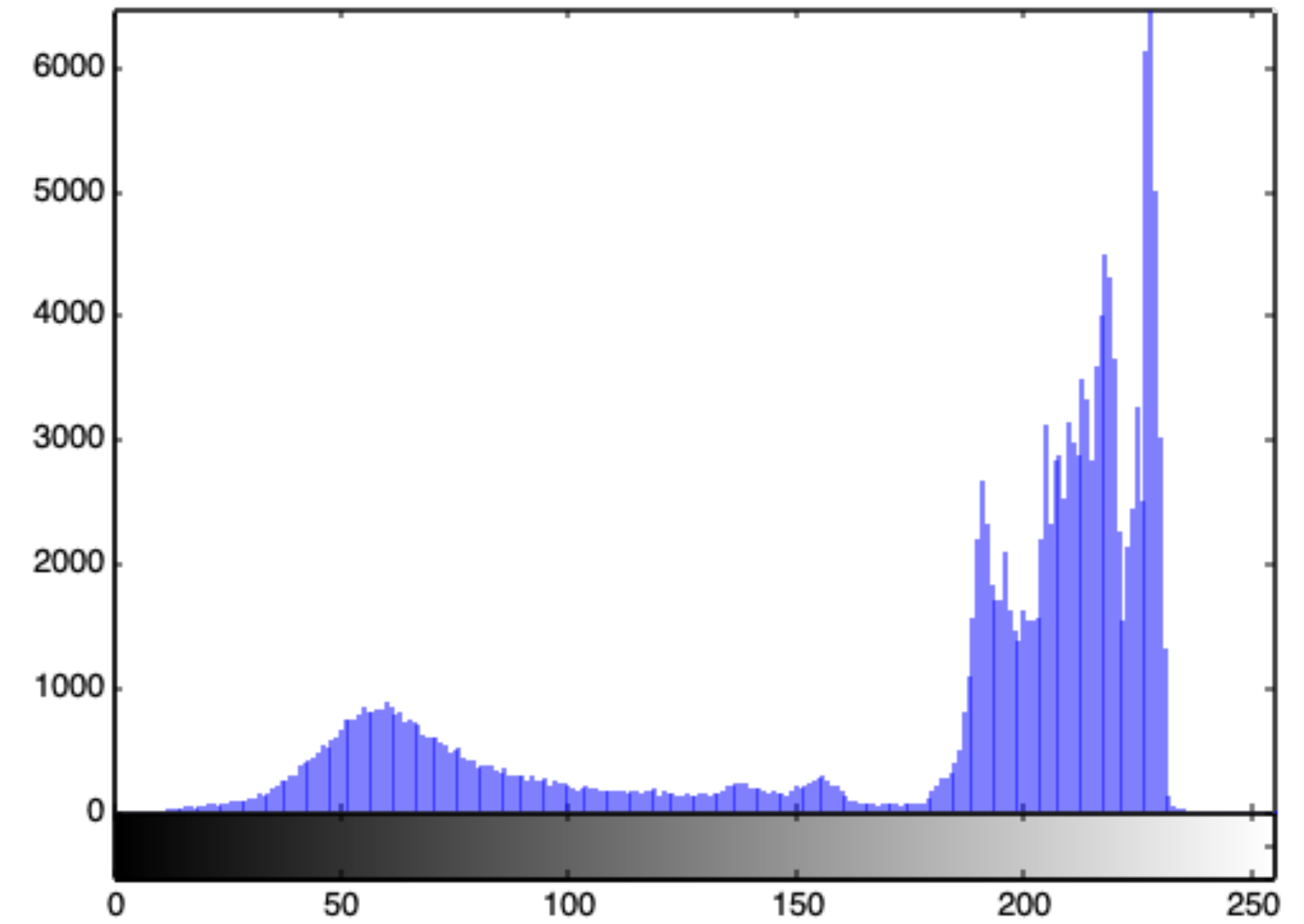
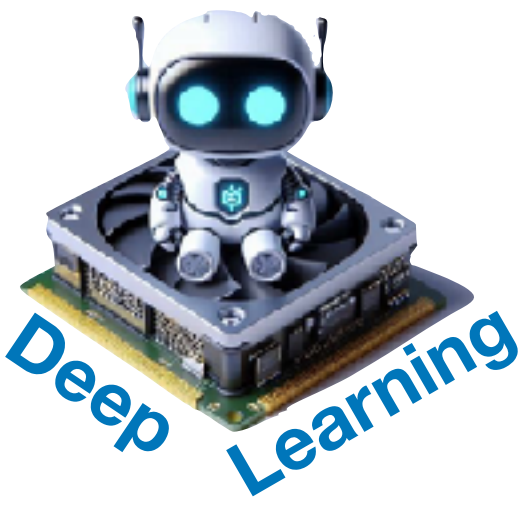


- This results in the quadratic equation:

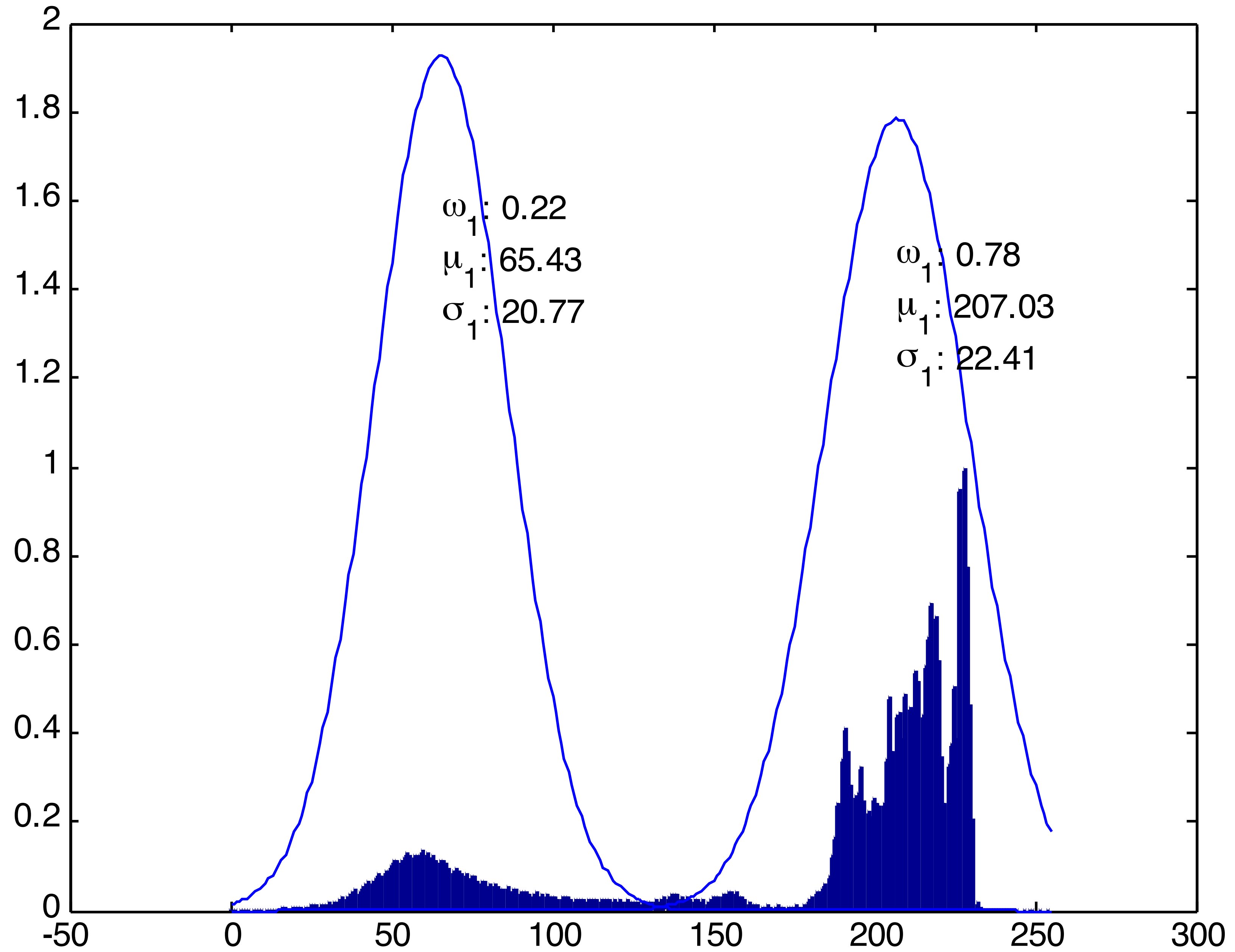
$$\left(-\frac{1}{2\sigma_1^2} + \frac{1}{2\sigma_2^2} \right) x_t^2 + \left(\frac{\mu_1}{\sigma_1^2} - \frac{\mu_2}{\sigma_2^2} \right) x_t + \left(-\frac{\mu_1^2}{2\sigma_1^2} + \frac{\mu_2^2}{2\sigma_2^2} - \ln \sigma_1 + \ln \sigma_2 \right) = 0$$

- The solution within range is taken...
 - It is possible to get two solutions
- Method is iterative, may converge to local maxima, provides MLE of the parameters

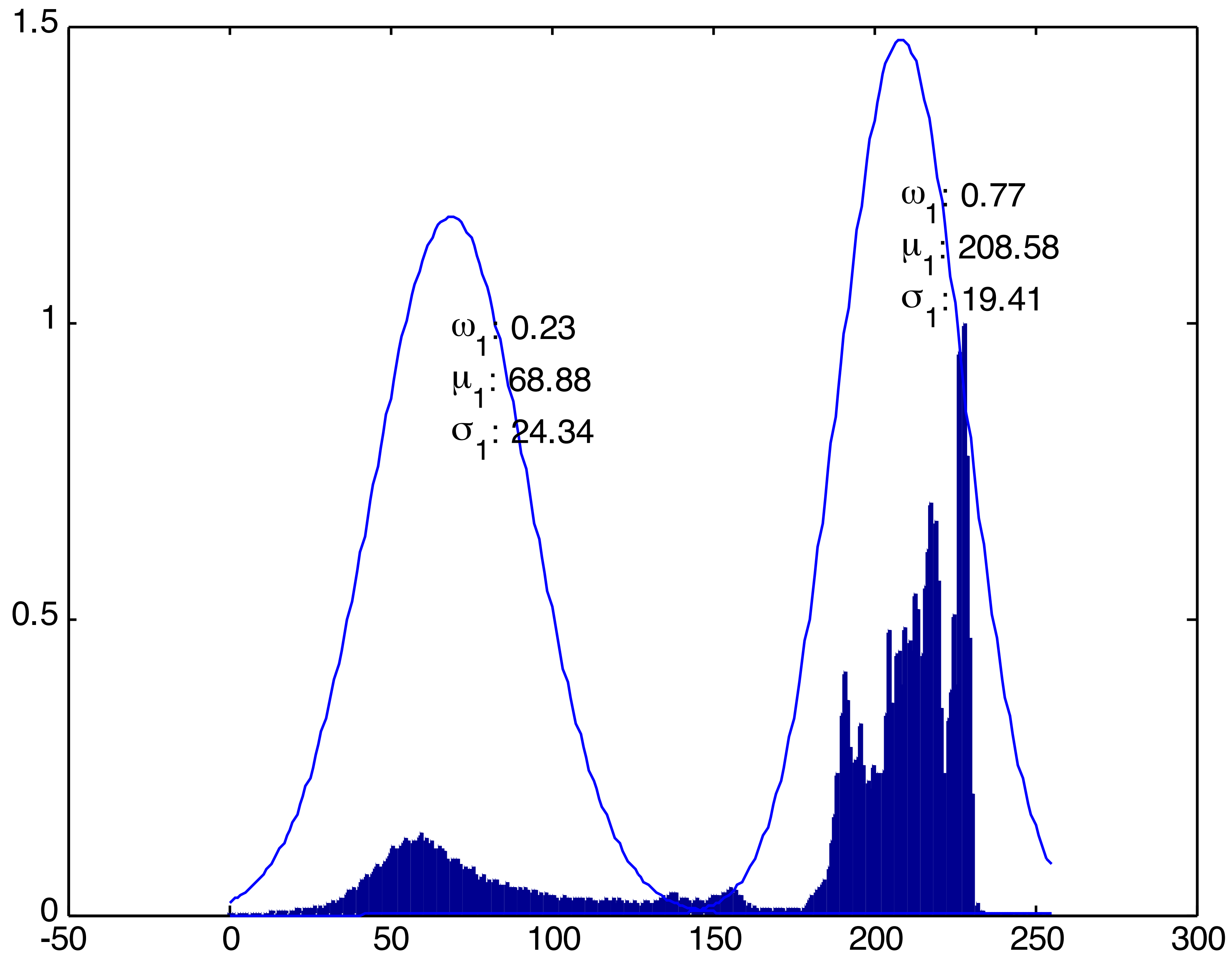
Automatic Thresholding Example



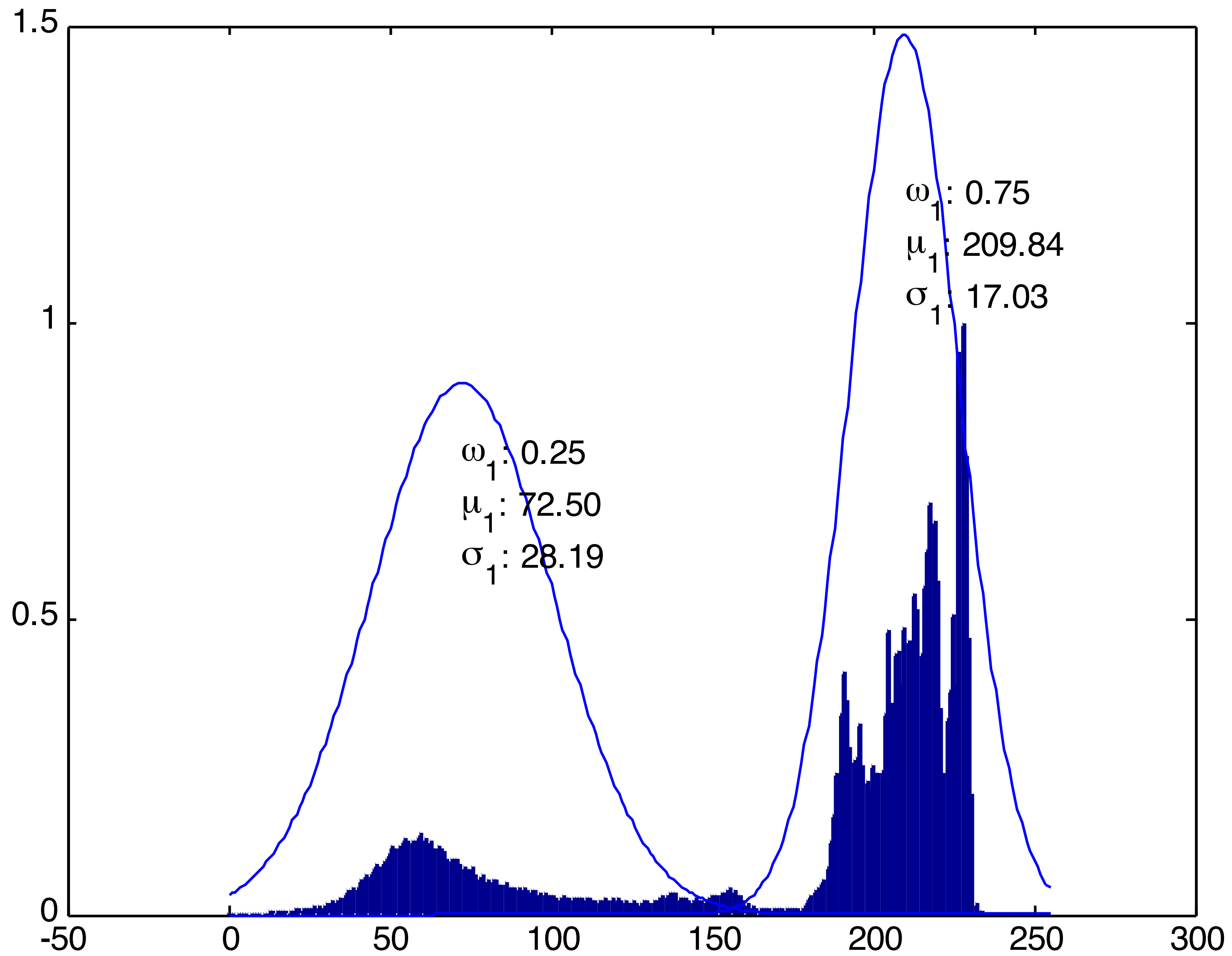
Iteration: 1



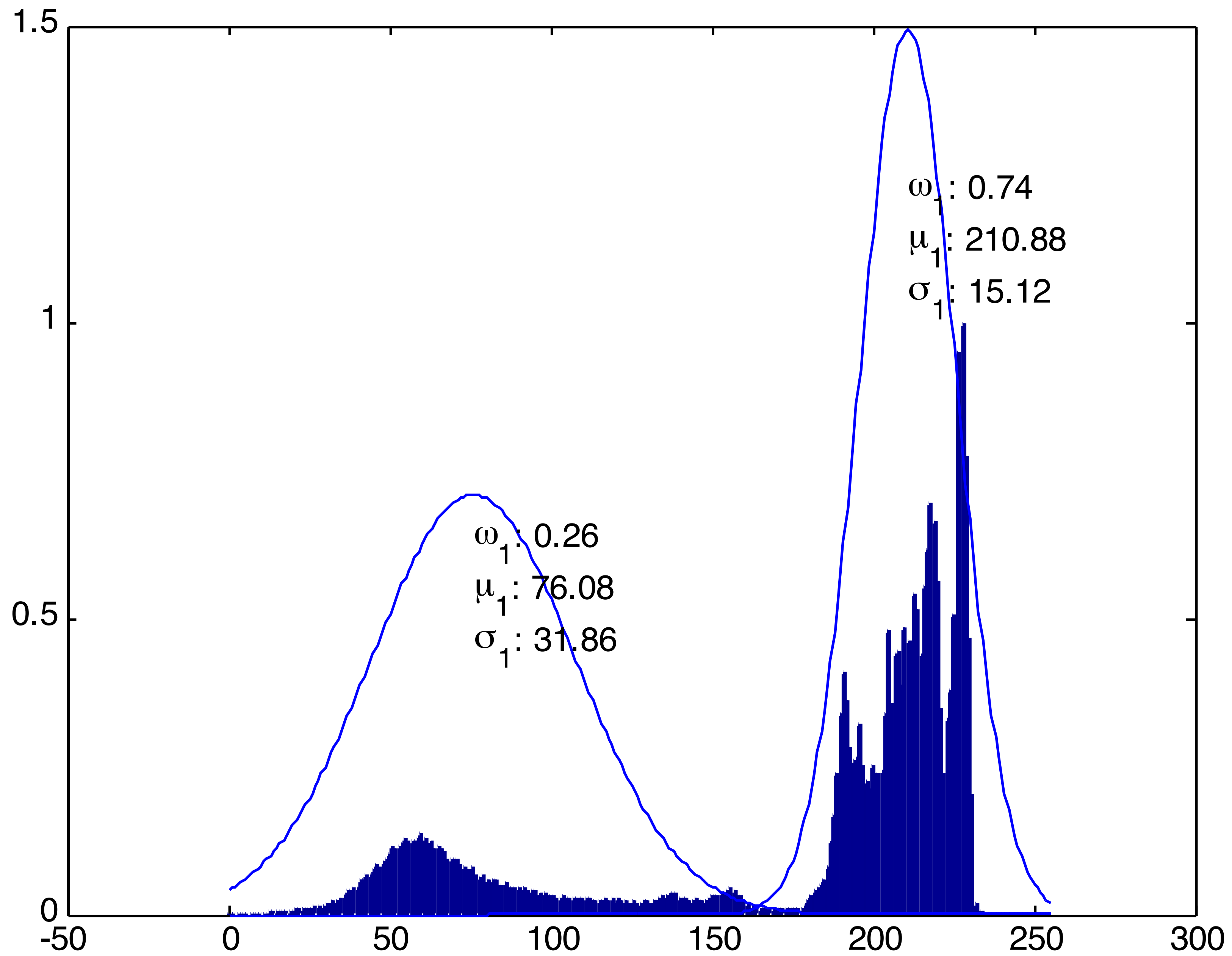
Iteration: 2



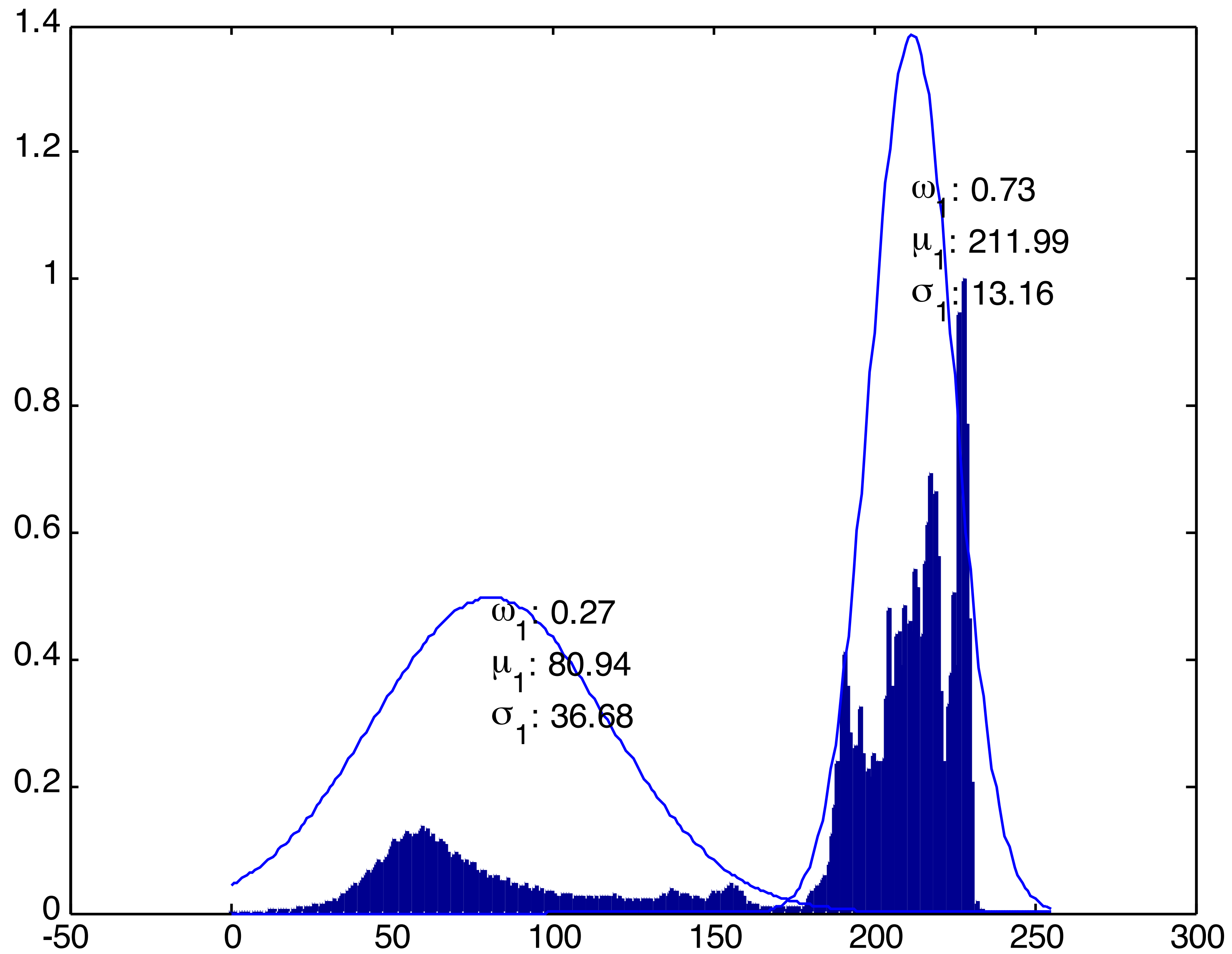
Iteration: 3



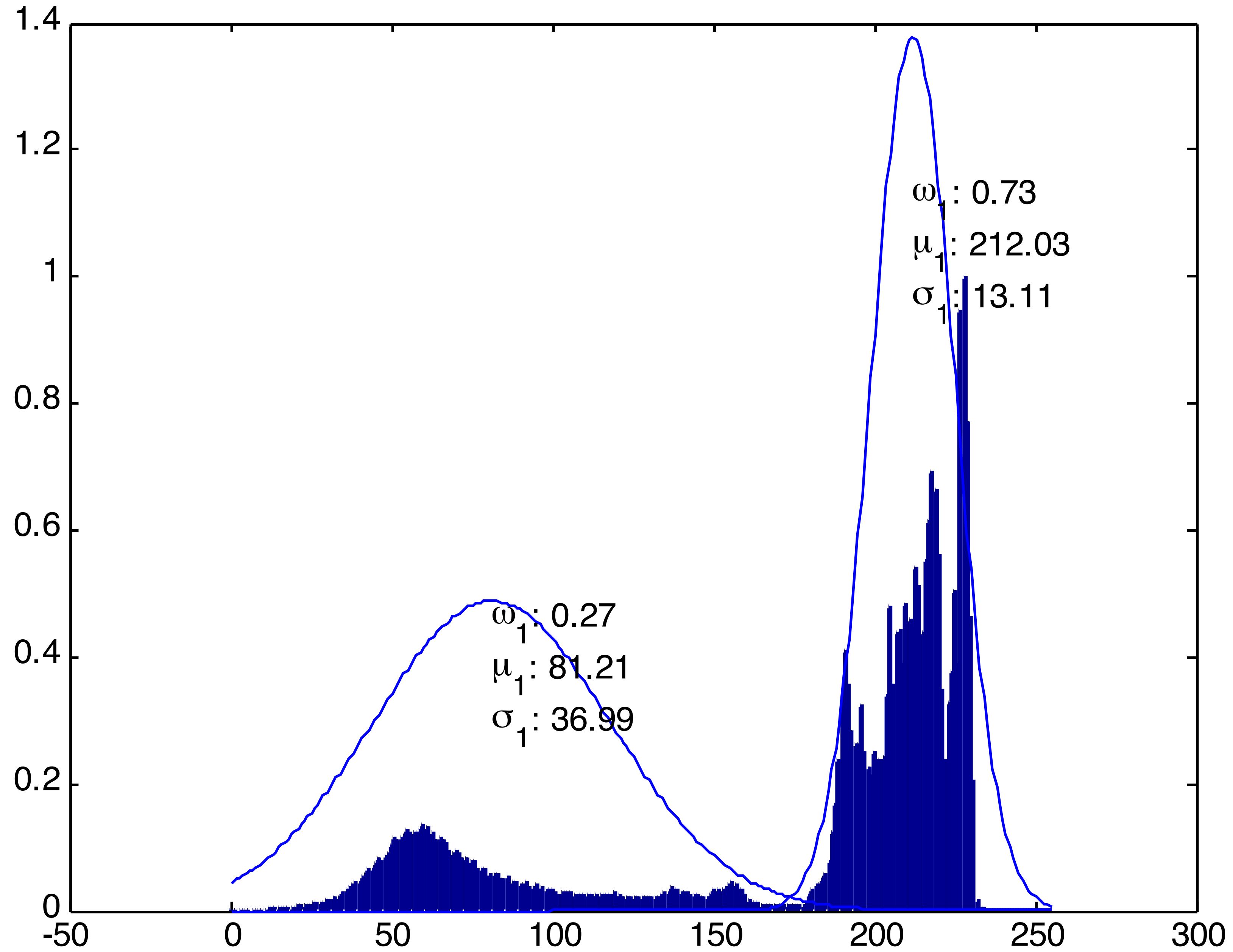
Iteration: 4



Iteration: 7



Iteration: 10



Input



After Segmentation



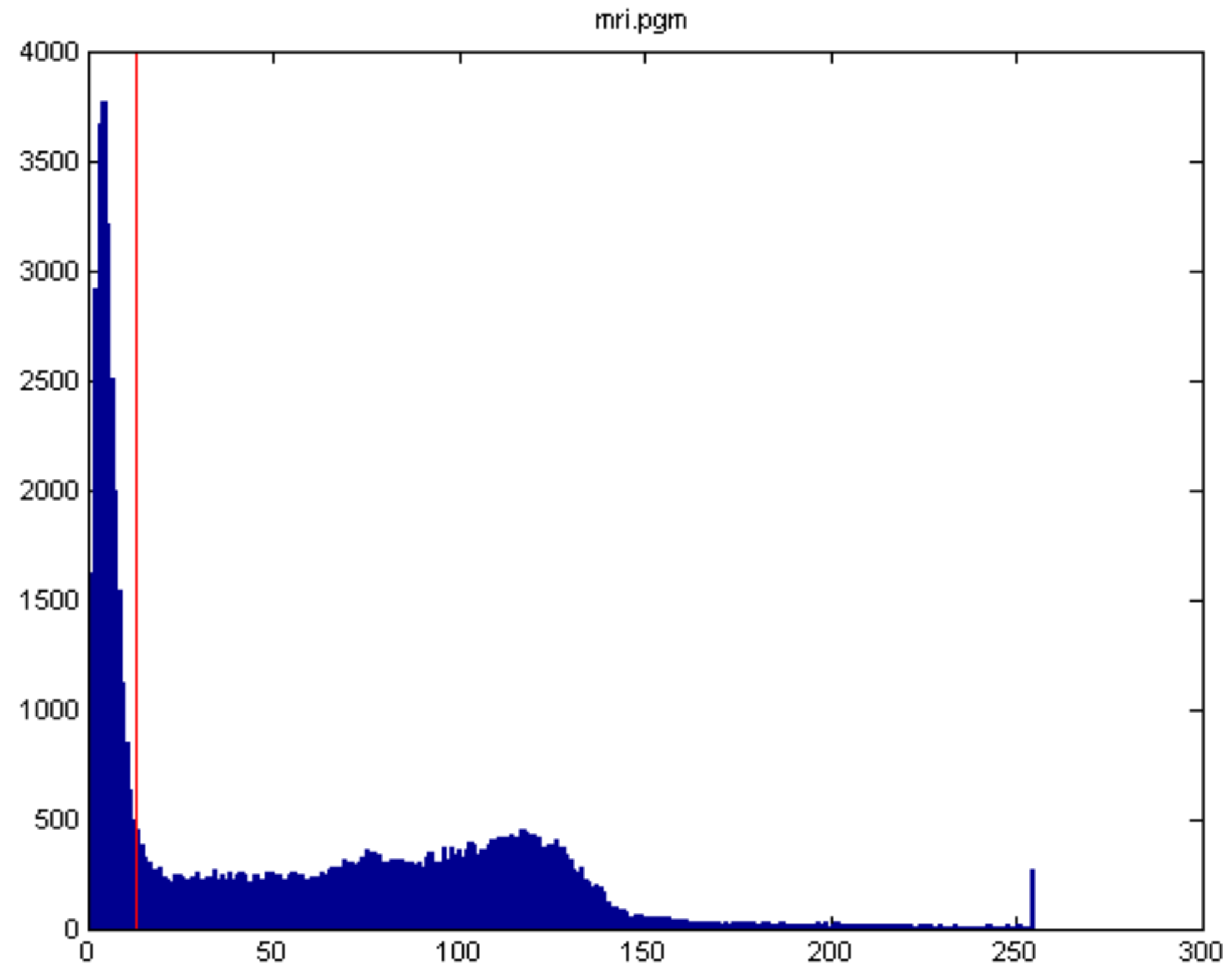
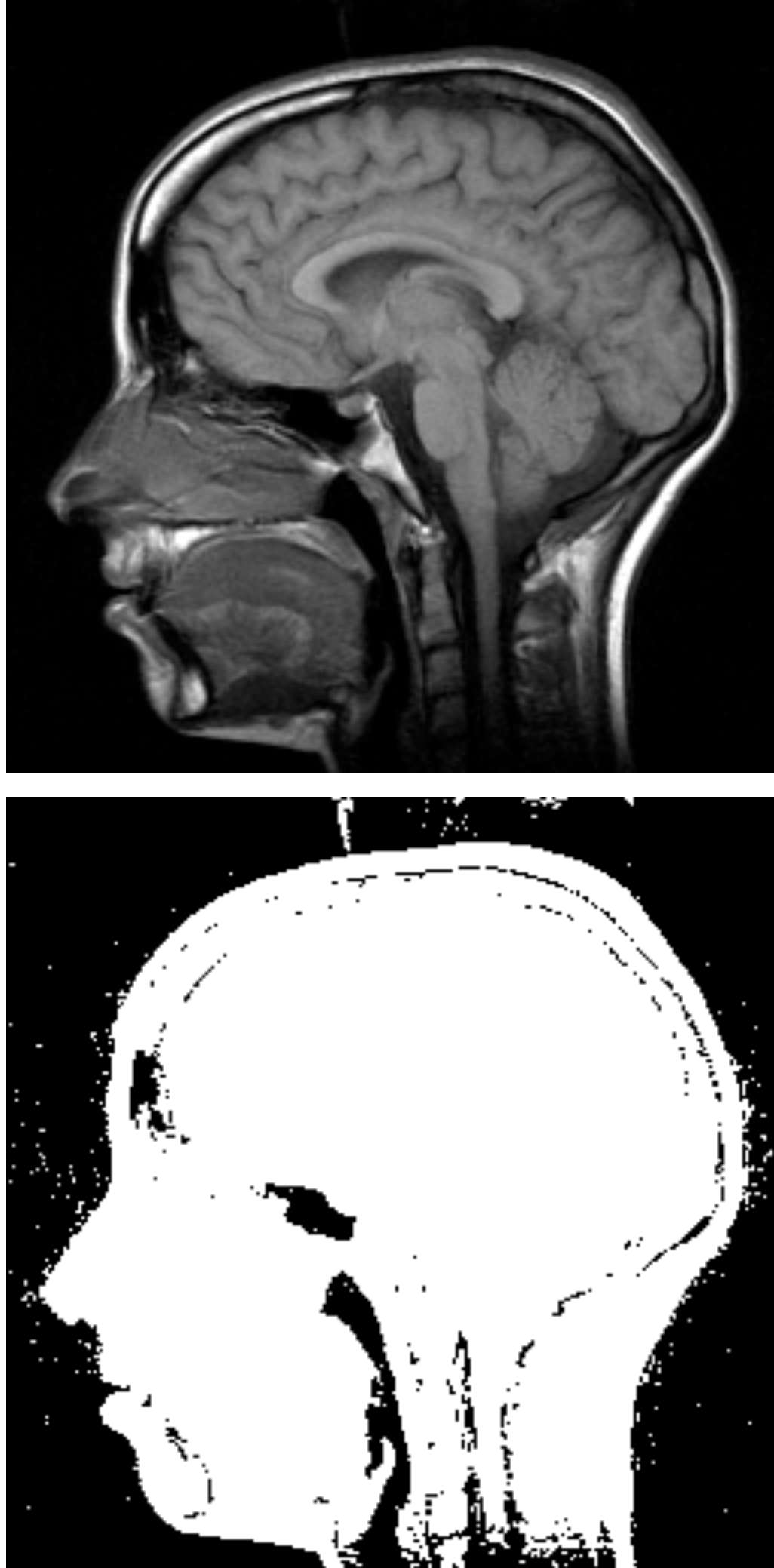
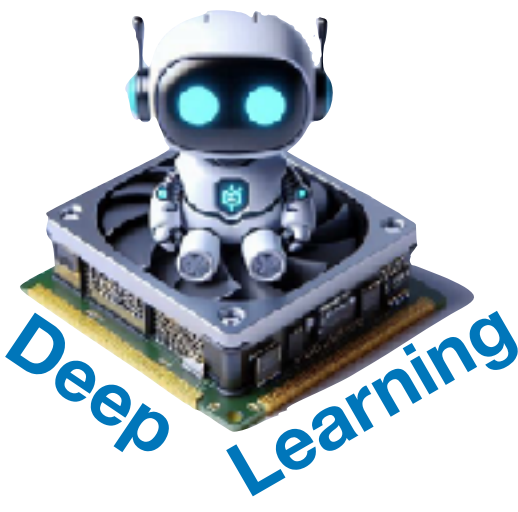
New background



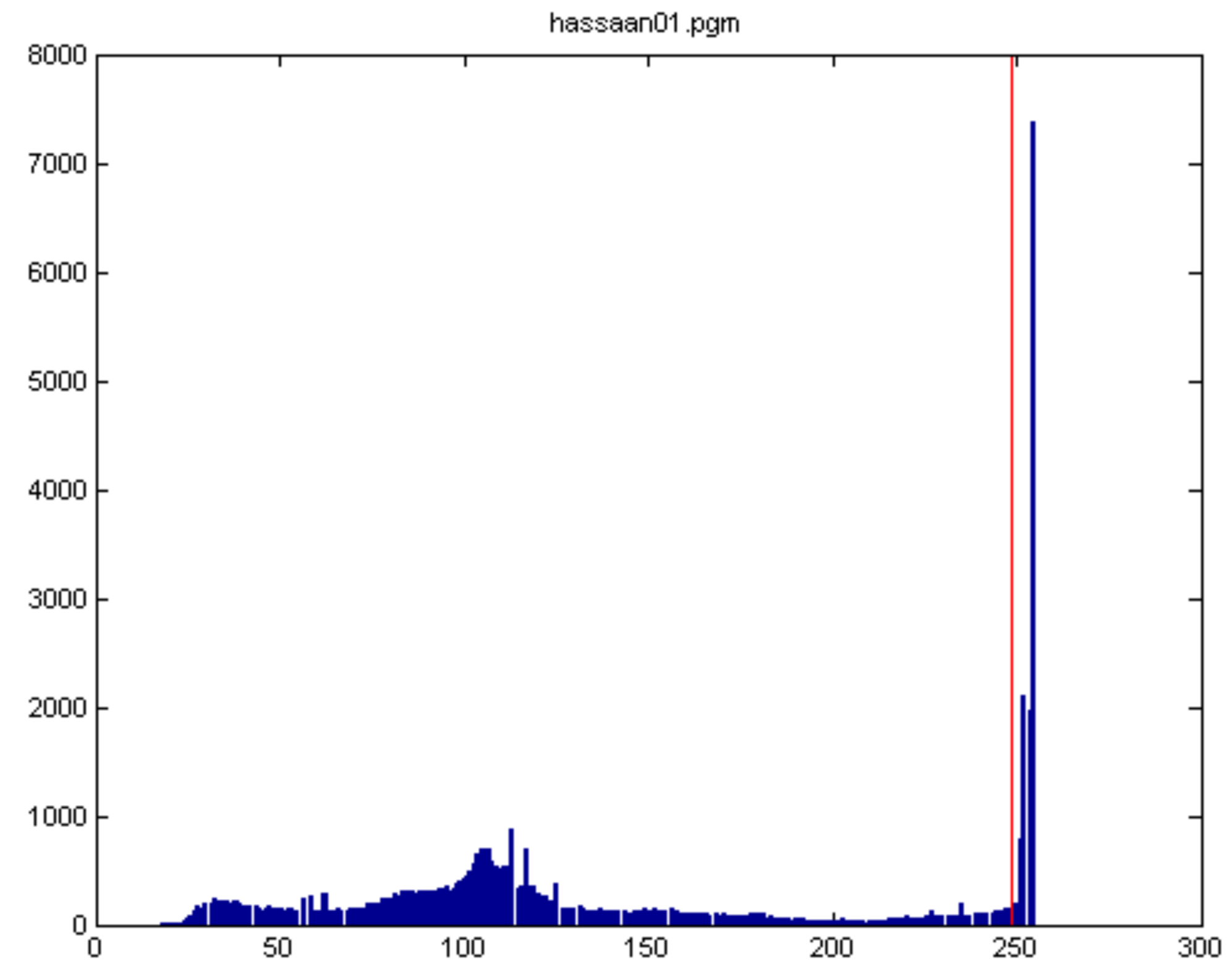
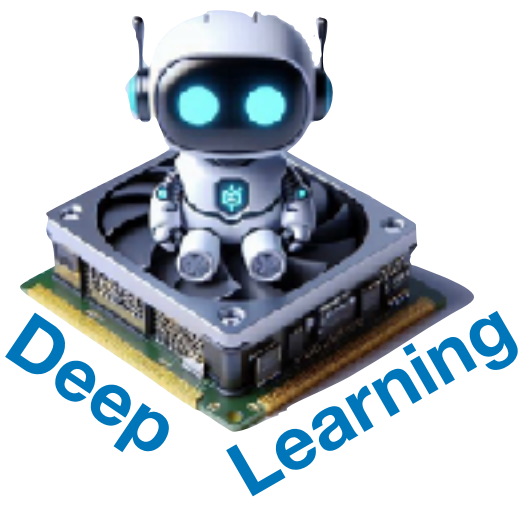
Final composite



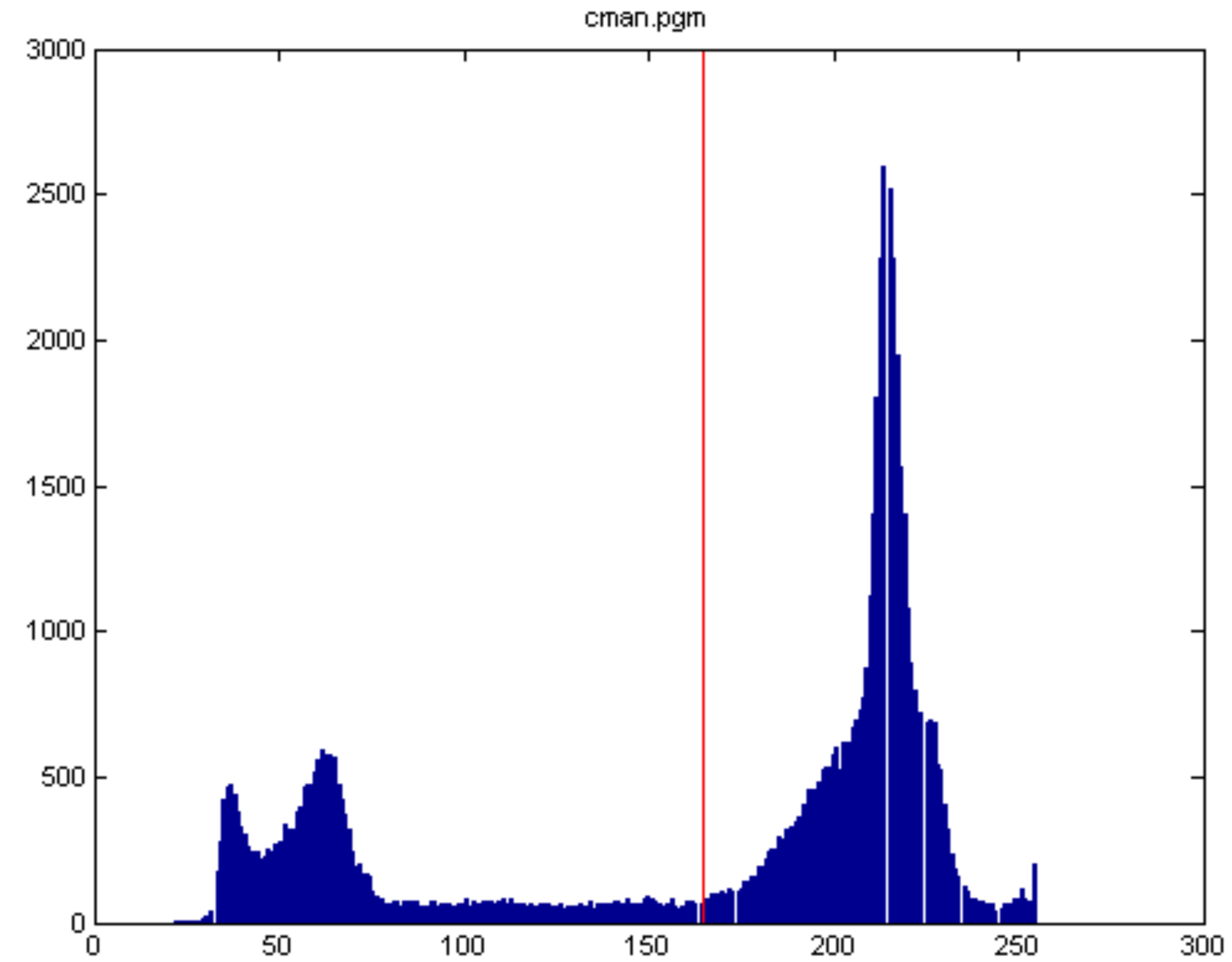
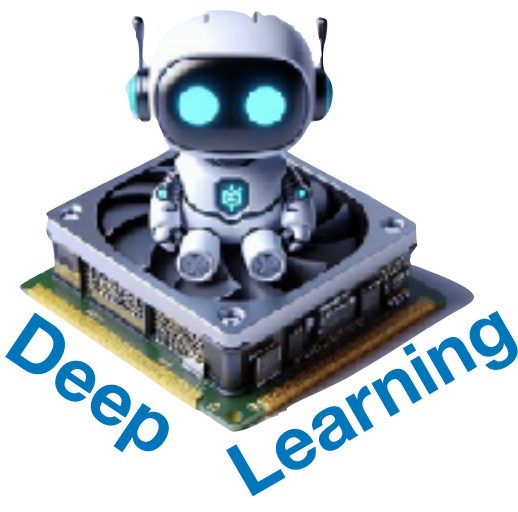
Automatic Thresholding Example



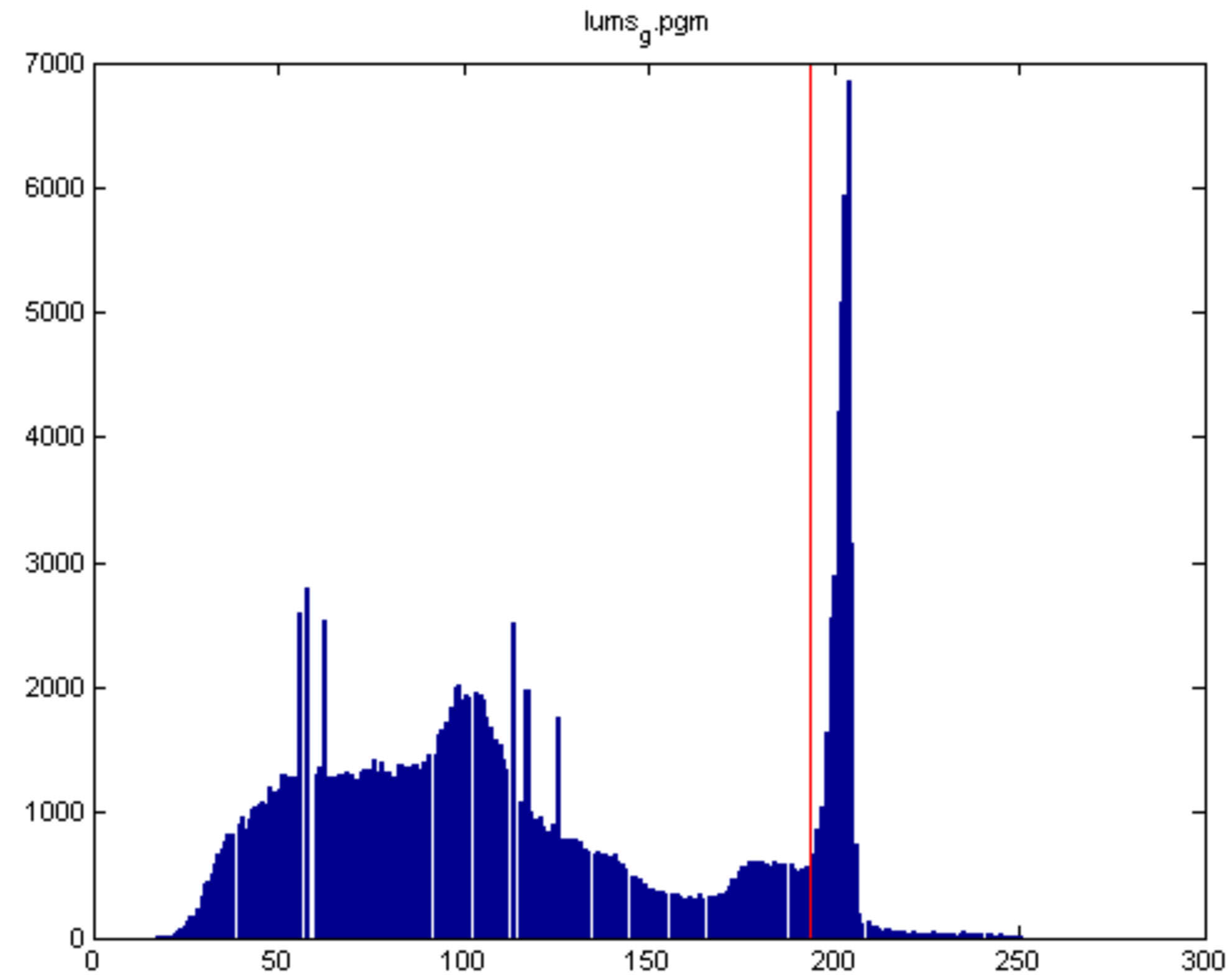
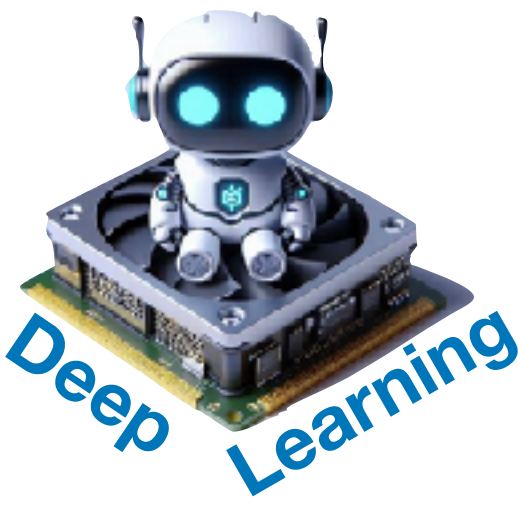
Automatic Thresholding Example



Automatic Thresholding Example

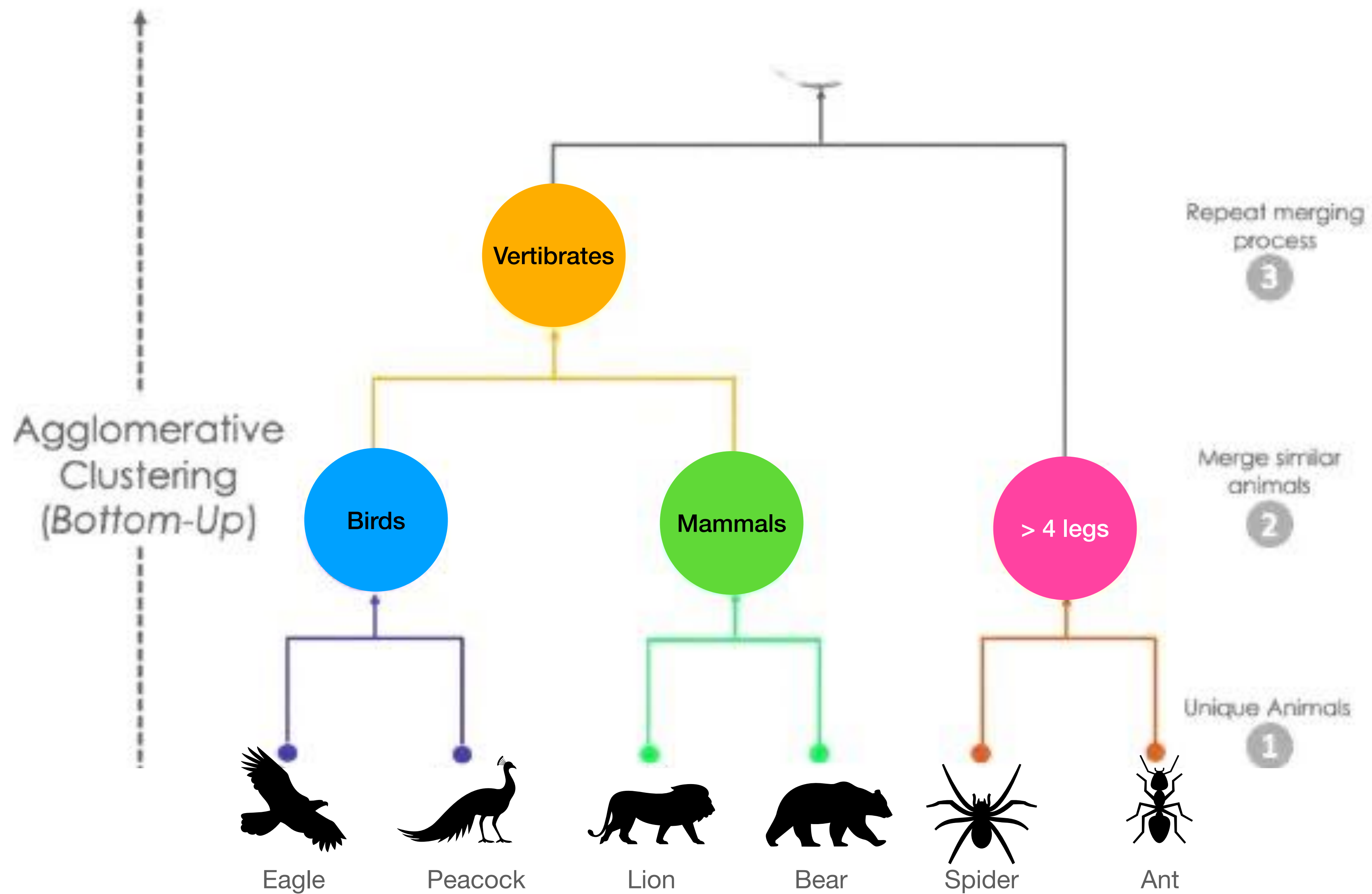


Automatic Thresholding Example



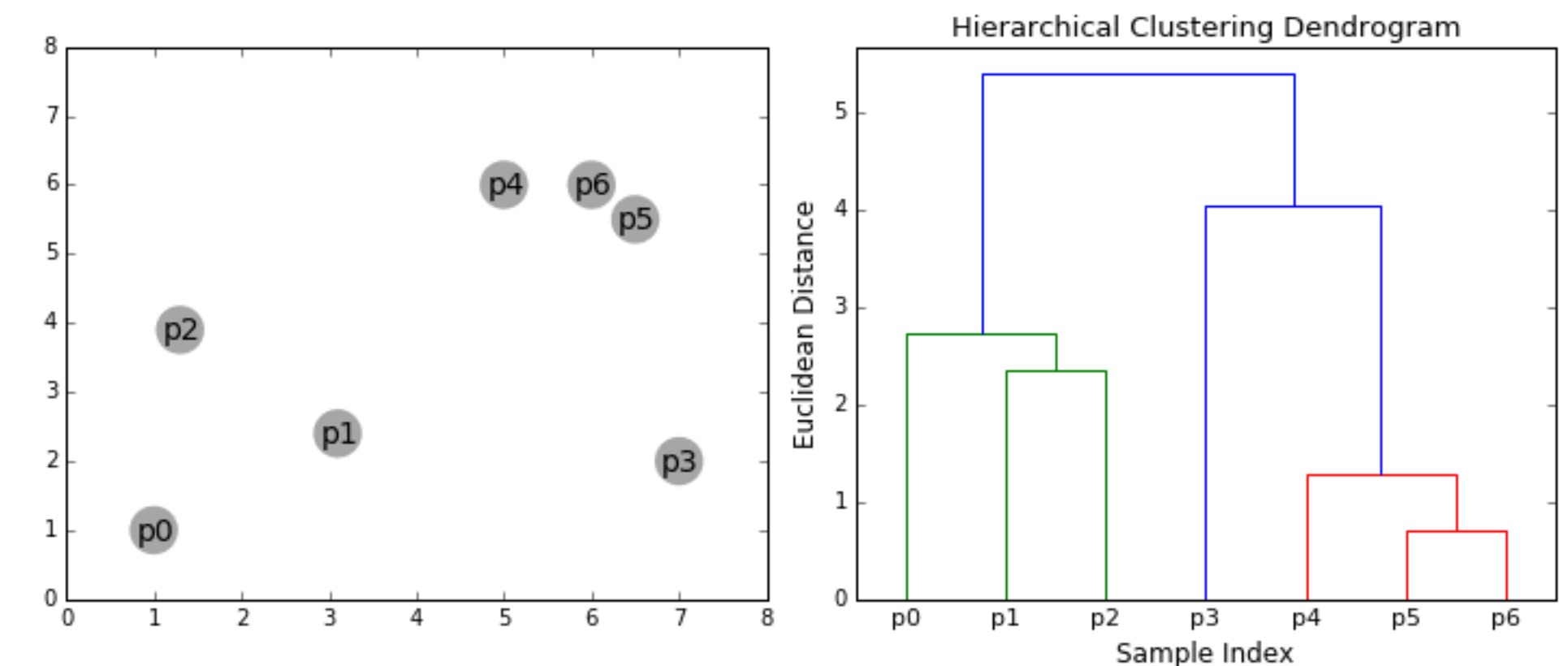
Hierarchical Clustering

The background features several overlapping, organic shapes in shades of pink and orange. A large, bright pink shape is on the left, partially overlapping a smaller, darker pink shape. To the right, there are more orange and pink shapes, some of which are semi-transparent, creating a layered effect. The overall aesthetic is modern and minimalist.



Hierarchical Clustering

- Creates a hierarchy of clusters (multi-level partitions)
 - returns a set of nested clusters
- Generally no requirement of a fixed number of k clusters
- Hierarchical method can be
 - **Divisive Approach (Top-Down)**
 - Initially all points are in **one huge cluster**
 - In every step one current cluster is **split into two**
 - Generates a top-down hierarchy of clusters
 - **Agglomerative Approach (Bottom-Up)**
 - Initially **each point** is a **cluster** itself
 - In every step **two clusters** are **merged** into one
 - Generates a bottom-up hierarchy of clusters



Hierarchical Clustering:

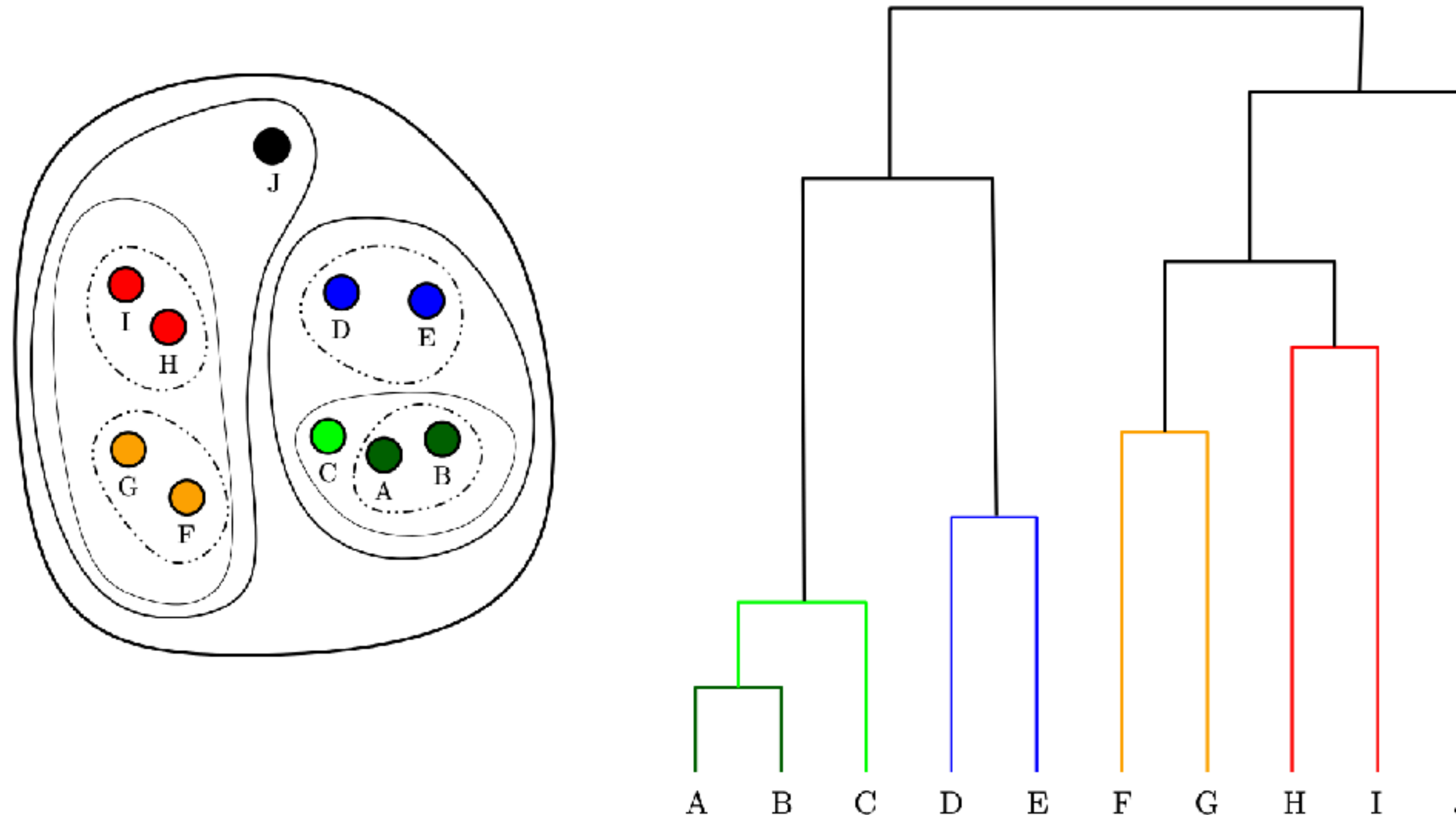
Agglomerative and Divisive



Hierarchical Clustering

Dendrogram

- Output of hierarchical clustering is represented by a **dendrogram** (a tree recording the sequence of merges or splits)



Agglomerative Clustering

Algorithm

Algorithm : Generic Agglomerative Clustering ($\mathcal{P}$)

Initialize with each point as a cluster in $\mathcal{C}$

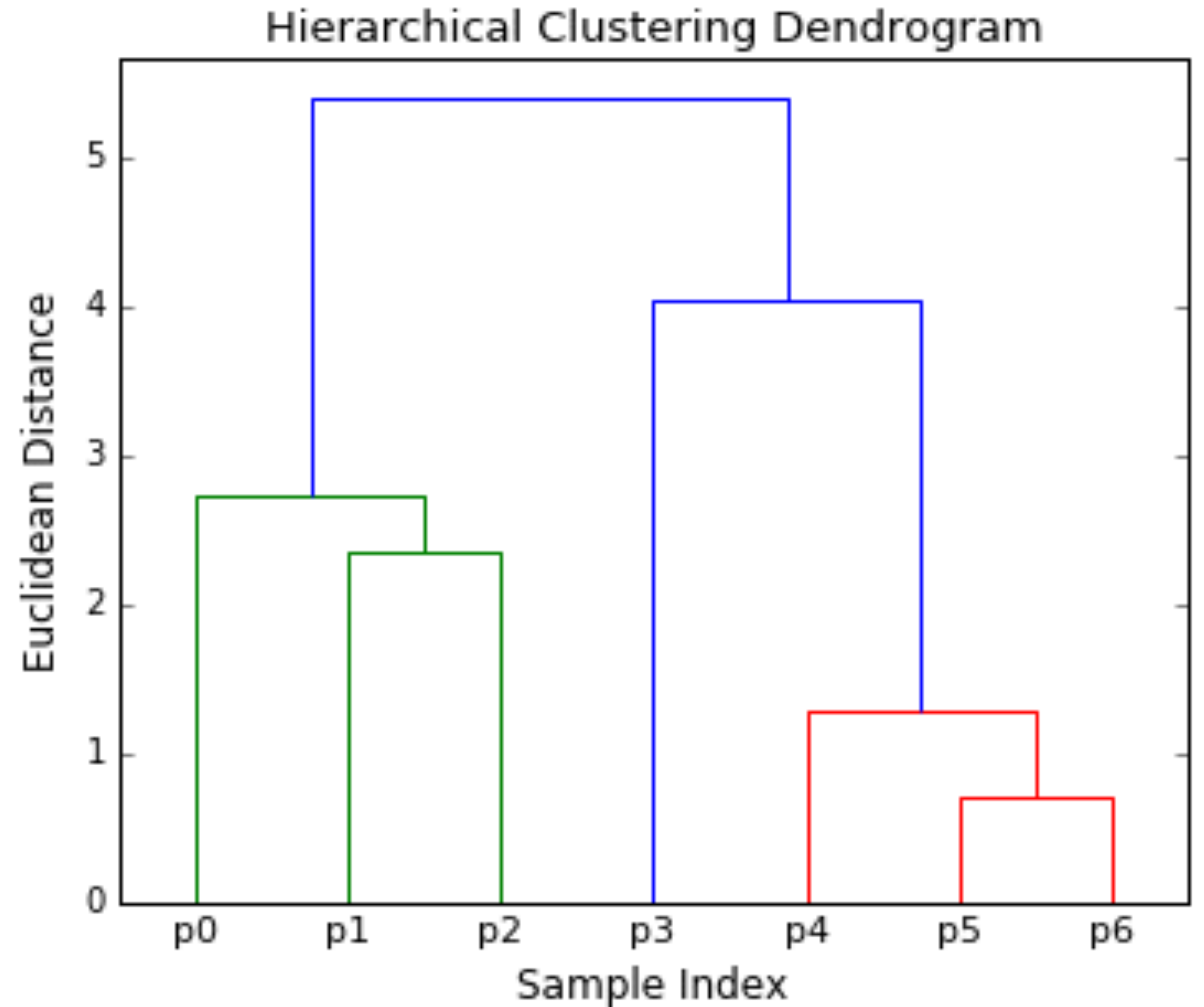
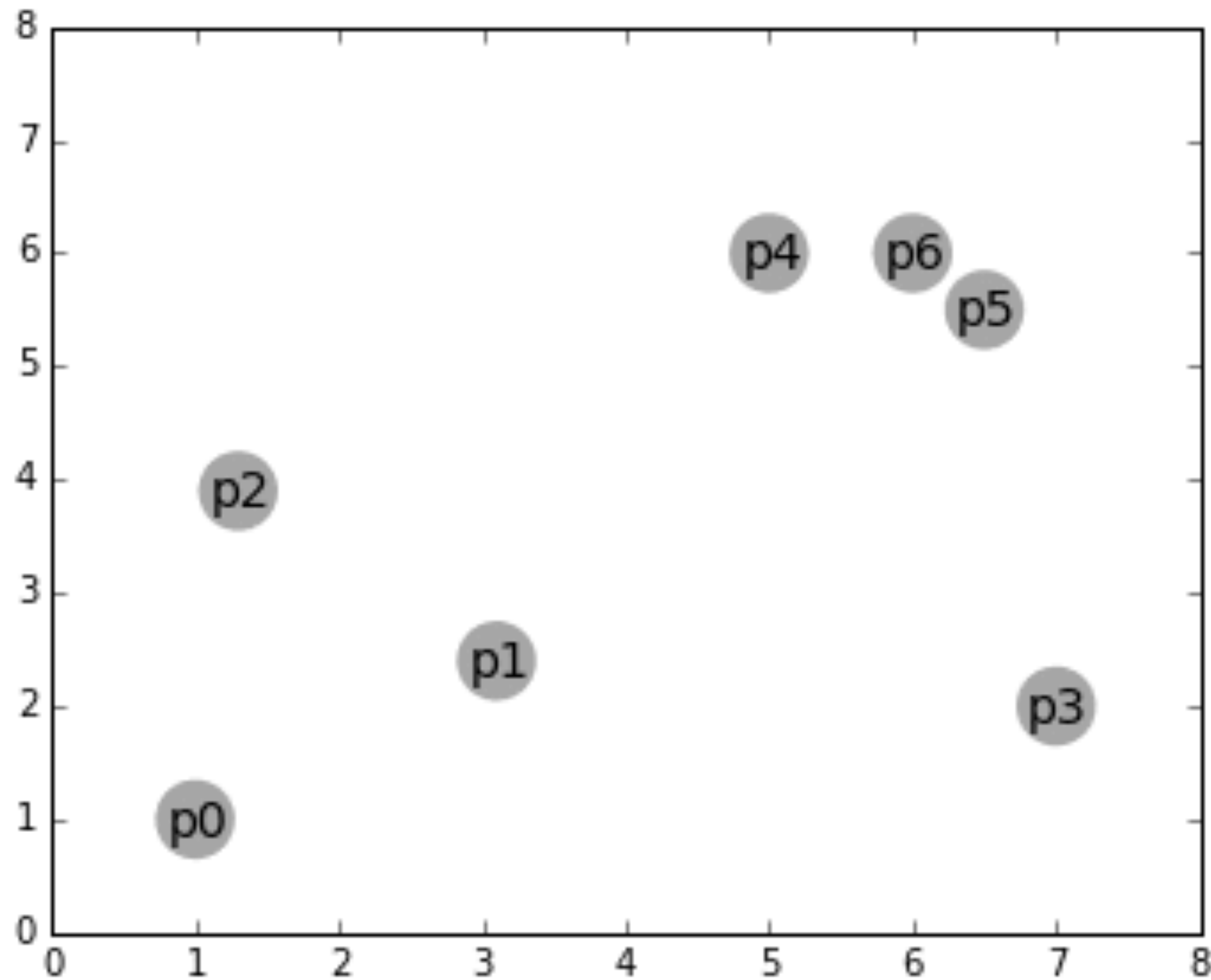
while **stopping criterion** is not met **do**

 Choose the **best pair** of clusters $(C_i, C_j) \in \binom{\mathcal{C}}{2}$

$C_m \leftarrow \text{MERGE}(C_i, C_j)$

$\mathcal{C} \leftarrow \mathcal{C} \setminus \{C_i, C_j\} \cup \{C_m\}$

Agglomerative Clustering



Hierarchical Clustering

(both divisive and agglomerative)

- is **rigid** in nature
 - once a cluster is made, it cannot be undone
 - less chances of improvement
- generally **less computational cost**
- does not require specific **number of clusters**
- Can stop when clustering is good enough (**stopping criterion**)

Hierarchical Clustering

Stopping Criteria

- **Diameter** of a cluster C , $\text{dia}(C)$ is the max inter-point distance in C
- **Radius** of a cluster C with centroid c , $\text{rad}(C)$ is the maximum distance of a point in C from the centroid c
- **Density** of a cluster C : is mass (number of points) over volume What is volume (shape) of the cluster? Use an estimate

Hierarchical Clustering

Stopping criteria based on quantity

- **Stop based on quantity:** **Stop** when the number of clusters is equal to k

Hierarchical Clustering

Stopping criteria based on ‘**quality**’

- **Stop** based on ‘**quality**’ of recently merged cluster, e.g. when the
 - **average inter-point distance** of the merged cluster is above a threshold
 - **diameter** of the merged cluster is above a threshold
 - **radius** of the merged cluster is above a threshold
 - the **average distance** from the centroid is above a threshold
 - **sum of squared distances** from the centroid is above a threshold

Hierarchical Clustering

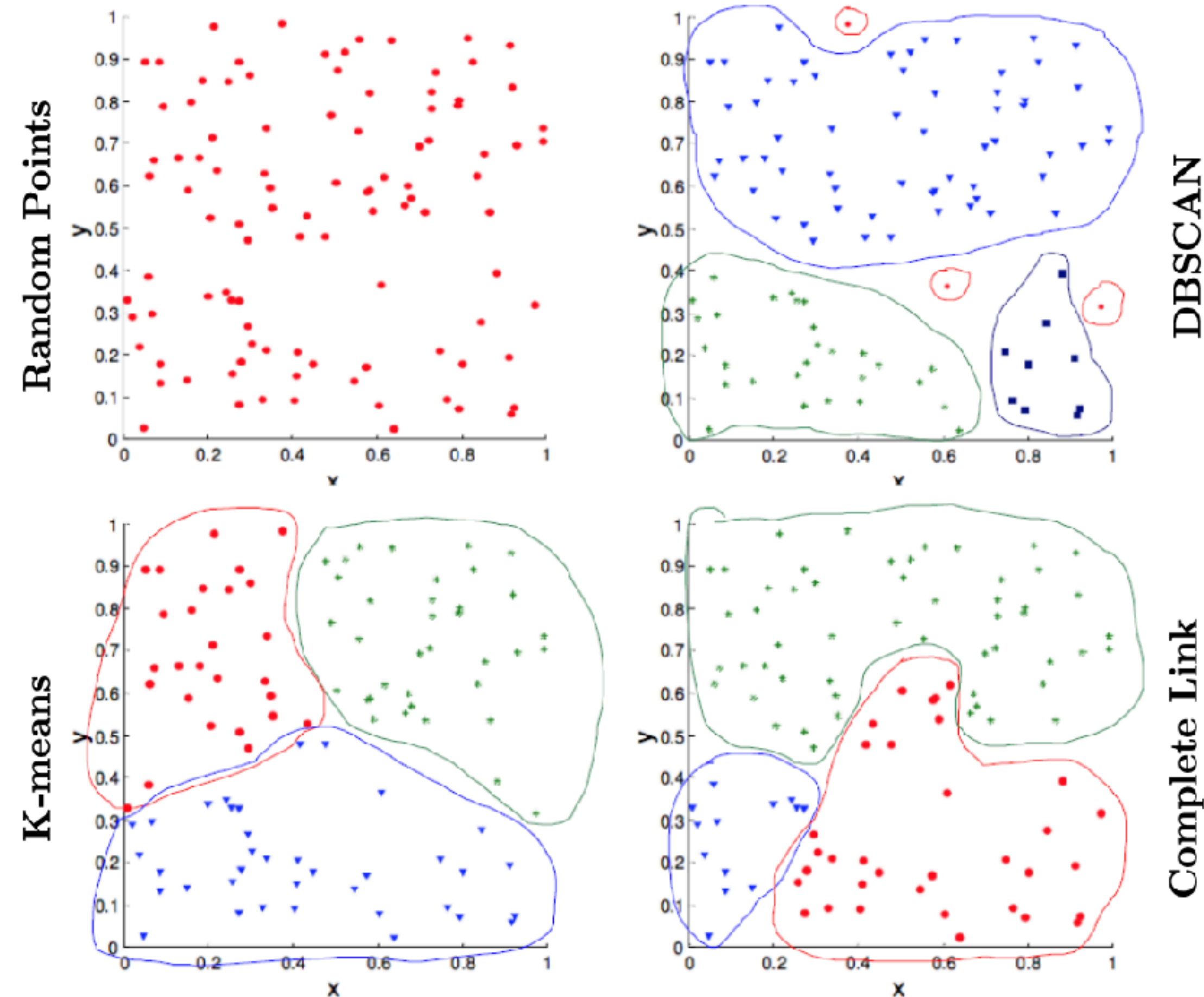
Stopping criteria based on a **global 'quality'**

- **Stop** based on a **global 'quality'** measure. e.g. when the
 - **average diameter** of all clusters increases above a threshold
 - the idea is as long as we merge cluster that truly should be merged, the **average diameter will not significantly increase**
 - when we merge a pair that should not be merged, there would be a **sudden jump in the average diameter**

Clustering Quality Assessment Validation and Evaluation

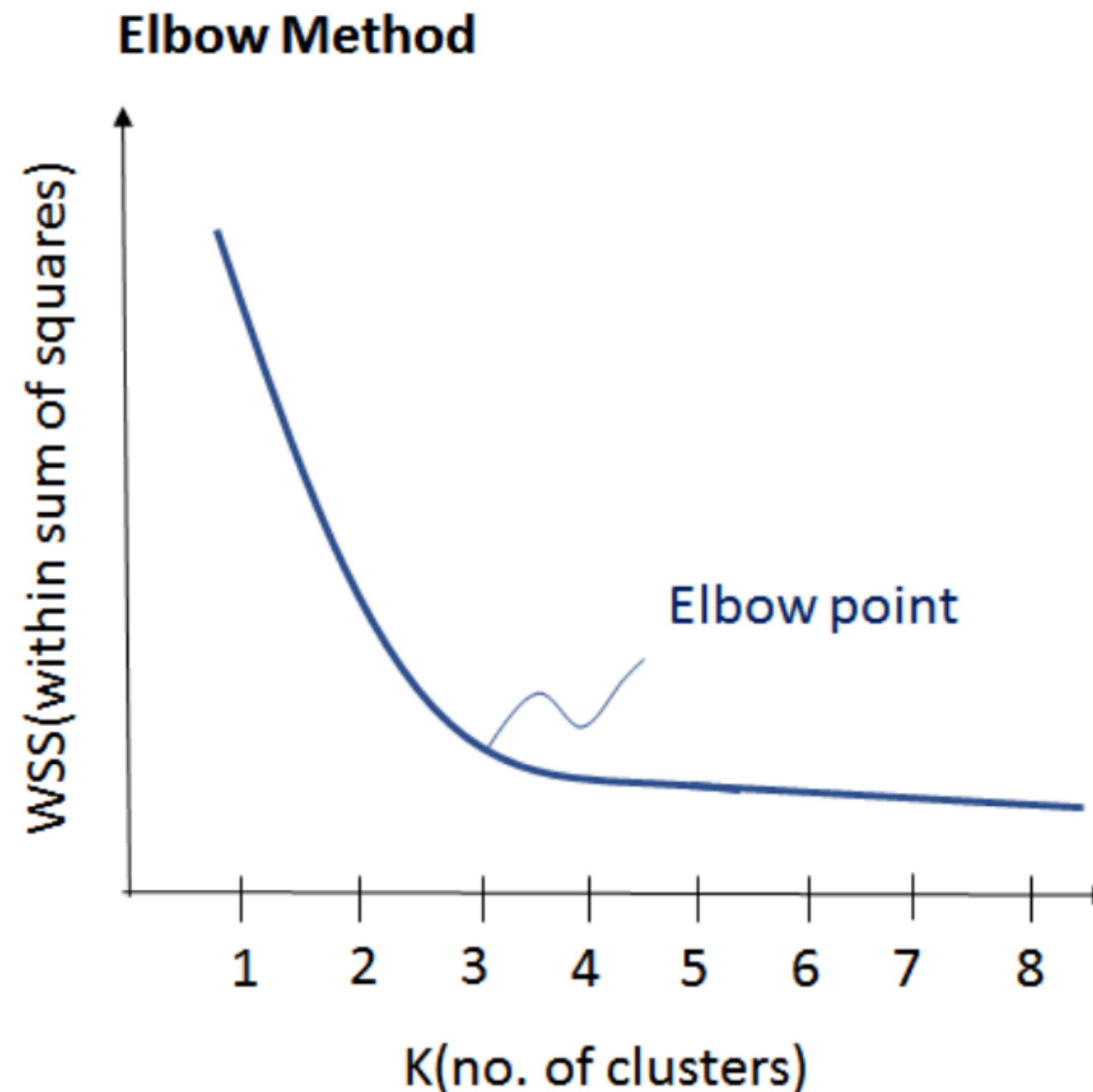
Goals & Aspects of Clustering Quality Assessment

Determine cluster tendency of dataset - meaningful groups



Goals & Aspects of Clustering Quality Assessment

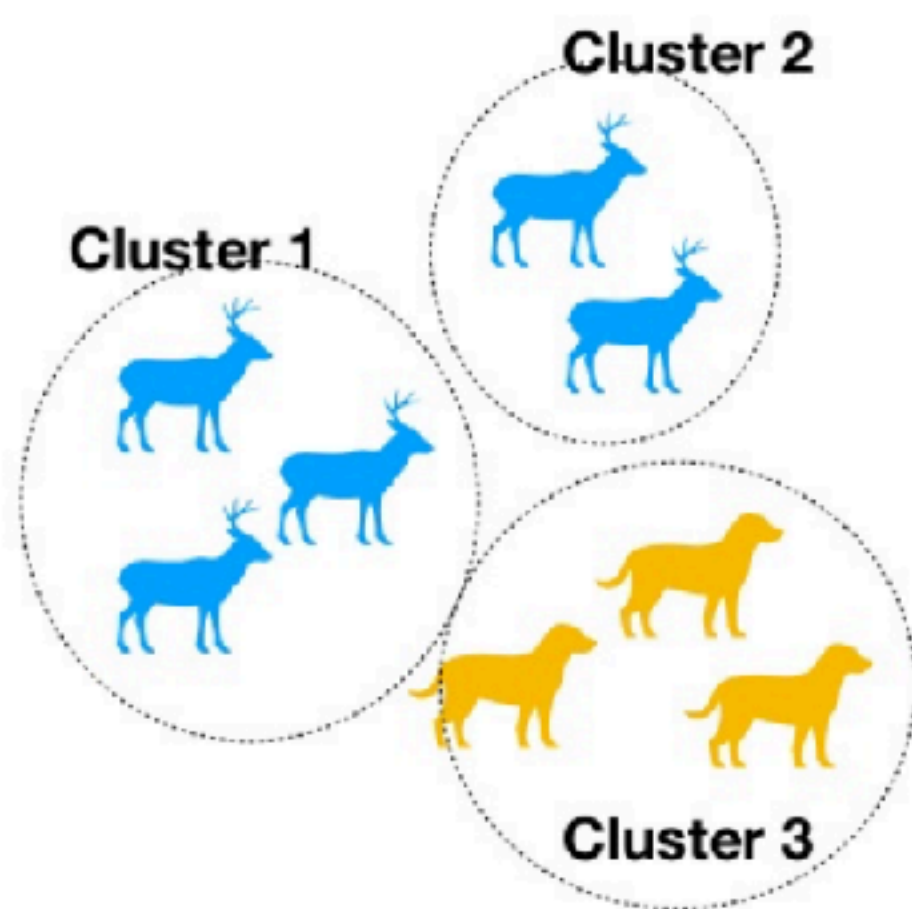
Find the correct number of clusters



Goals & Aspects of Clustering Quality Assessment

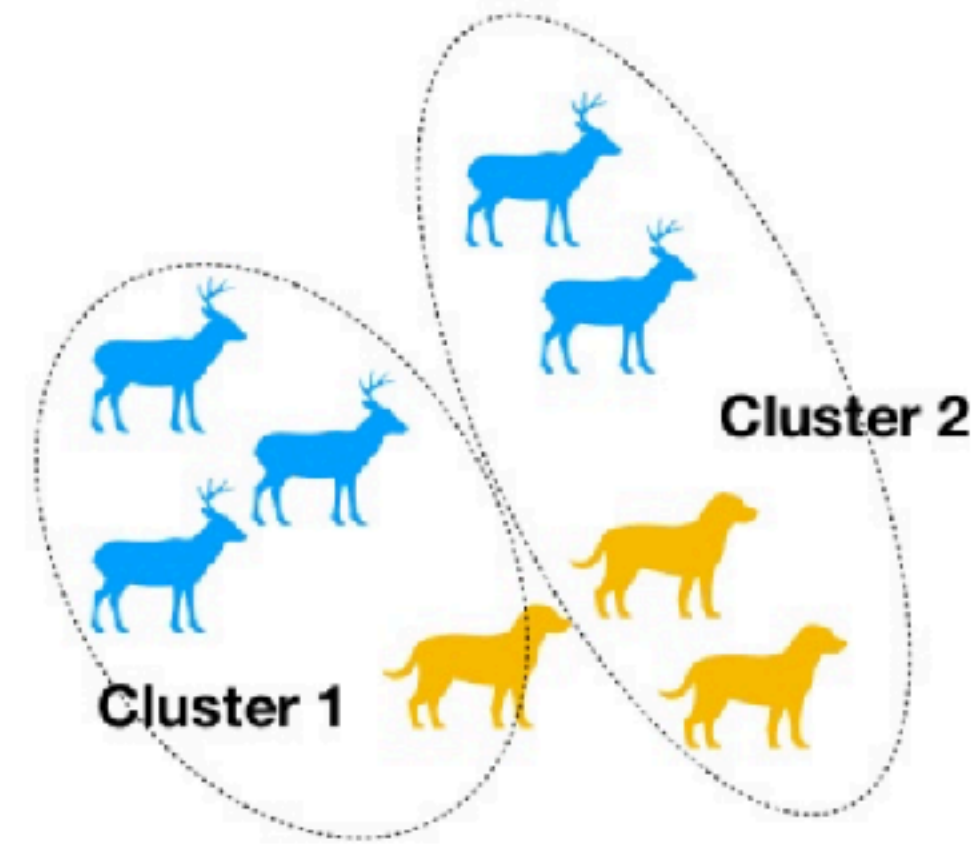
Validate Clustering Quality

- Clustering is an unsupervised task (cannot use ground truth in the clustering algorithm)

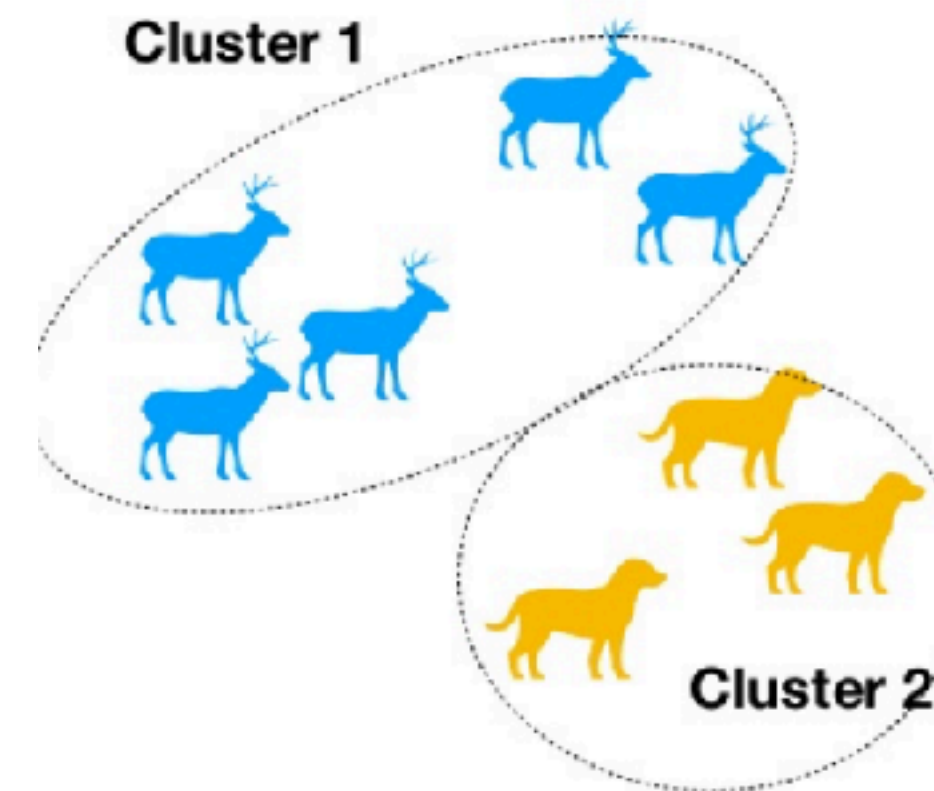


Good

Homogeneity

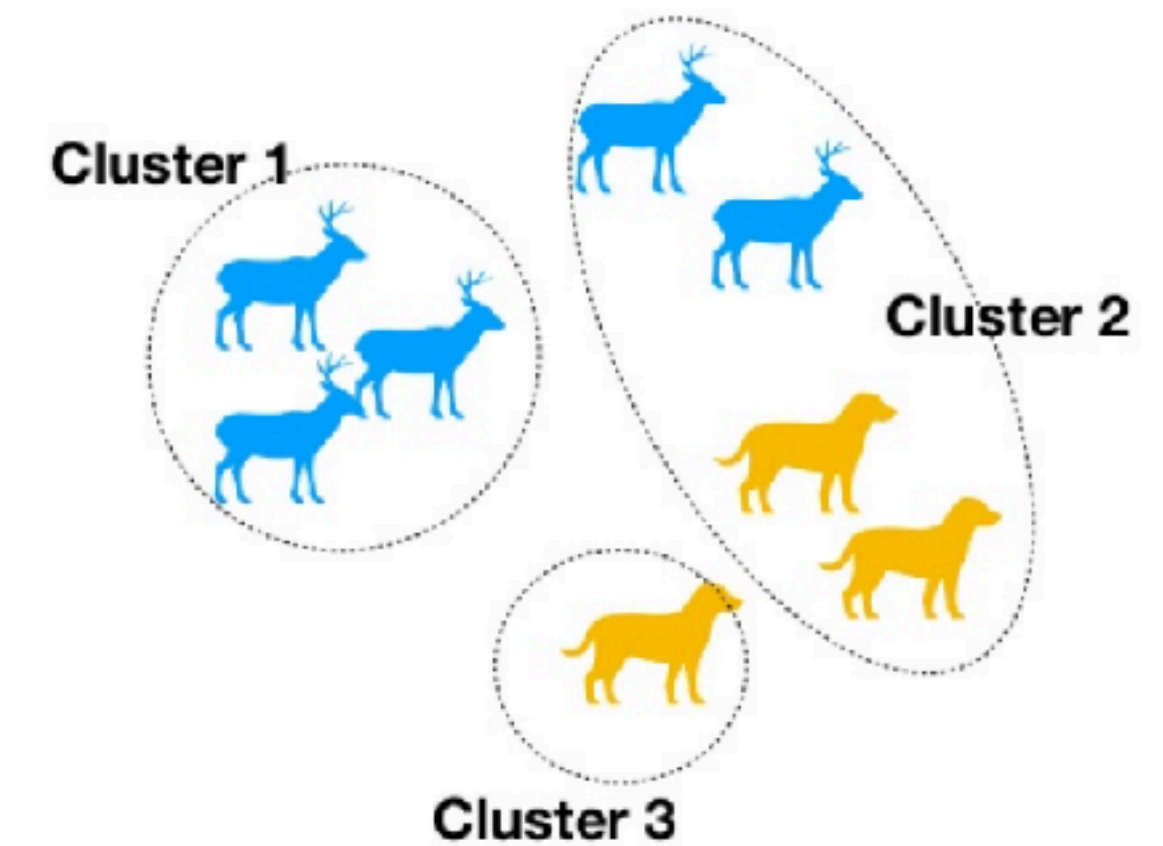


Bad



Good

Completeness

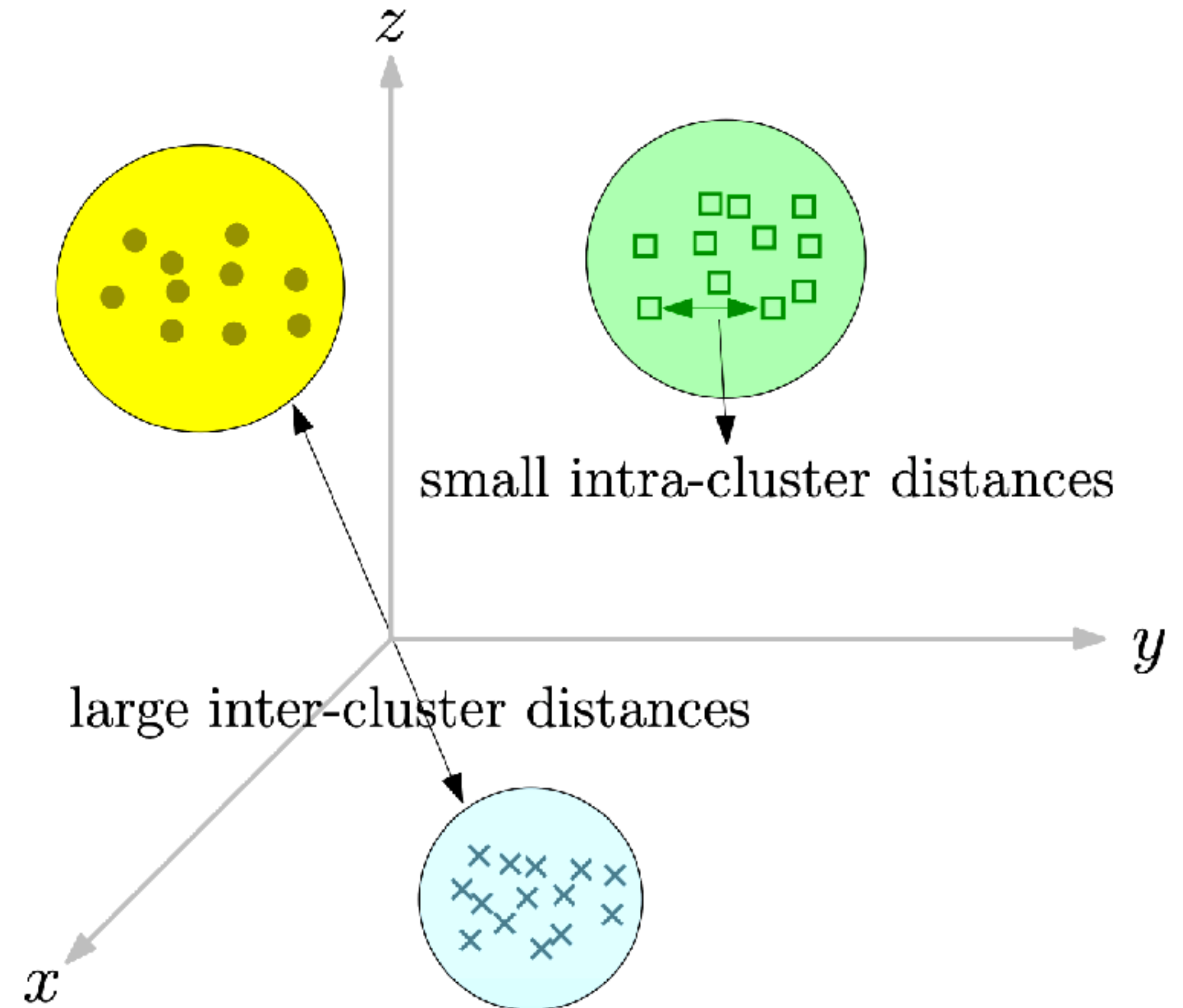


Bad

Goals & Aspects of Clustering Quality Assessment

Cluster Compactness & Separation

- Intra-Cluster-Low distances
- Low Intra-Cluster variance
- Inter-cluster high distances



Goals & Aspects of Clustering Quality Assessment

External Measures

- Any measures for assessment of quality of classification can be used
- Accuracy
- Error
- Precision,
- Recall,
- F1 measure
- Purity Entropy
- Conditional Entropy
- Normalized Mutual Information (NMI)
- Maximum Matching: Match clusters to class and see goodness of matching

Next ...

Decision Trees

